Statistical Foundations

for Quantitative Work

Answer Key

Worked solutions to the end-of-chapter exercises

Each solution is numbered to match its exercise in the book. Numerical answers were computed at full precision and rounded for display, so a hand calculation may differ in the last digit. Answers to exercises marked *Computer* come from simulation or software and will vary slightly with the random seed.

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Chapter 1

Why Statistics?

Solution 1.1 (Descriptive or inferential?)**.**

(a) Descriptive. It reports a property of the 250 numbers in hand and is simply true of them.

(b) Inferential. Next winter has not happened; the statement generalizes from observed data to unobserved outcomes.

(c) Descriptive. It summarizes the 1,200 responses actually received.

(d) Inferential. It extends the sample result to all customers, most of whom were not surveyed. (It also quietly assumes the respondents are representative, which Exercise 1.6 questions.)

(e) Descriptive. It is a fact about the records in the log.

(f) Inferential. The first clause is descriptive, but the conclusion is a claim about next year, and it rests on the assumption that the process generating returns is stable from one year to the next.

The test in every case is whether the claim stays inside the data or reaches beyond it.

Solution 1.2 (Parameters and statistics)**.**

(a) Parameter, the population mean μ.

(b) Statistic, the sample mean $\bar{x} = 31.7$.

(c) Statistic, the sample standard deviation $s = 4.2$.

(d) Parameter, a population proportion, usually written p.

(e) Statistic, the sample proportion $\hat{p} = 0.12$.

(f) Only the statistics in (b), (c) and (e). The parameters are fixed features of the year's 8,760 hours; the statistics depend on which 400 hours happened to be drawn. That dependence is sampling variability, the subject of Chapter 6.

Solution 1.3 (Types of data)**.**

(a) Discrete numerical (a count). The mean is meaningful: "4.3 trades per hour on average" is a sensible statement even though no single hour has 4.3 trades.

(b) Continuous numerical. The mean is meaningful.

(c) Ordinal. The categories are ordered, but the gap from AAA to AA is not the same kind of quantity as the gap from BB to B, so averaging codes such as $1, 2, 3, \ldots$ is not meaningful. Report the distribution across categories, the median category, or default rates by category.

(d) Nominal. No mean; report counts or the modal region.

(e) Nominal, despite being written with digits. The digits are labels; an "average postal code" is meaningless. This is the classic case of a categorical variable disguised as a number.

(f) Continuous numerical: a duration, recorded here to the nearest minute but continuous in principle. The mean is meaningful, though durations are usually right-skewed, so report the median as well (Chapter 2).

(g) Ordinal. Averages of such scores are reported constantly, but they assume the steps $1 \to 2$ and $4 \to 5$ are equal, which nothing guarantees. The median and the full distribution of responses are the defensible summaries; a mean is at best a rough index.

(h) Nominal. No mean; report shares of each type.

Solution 1.4 (Cross-sectional or time series?)**.** (a) and (c) are time series: one unit observed repeatedly over time. (b) and (d) are cross-sectional: many units observed at essentially one time.

Dependence is expected in the time series. Today's price is close to yesterday's, and this hour's load is close to last hour's. Such dependence (autocorrelation) means that 8,760 hourly loads carry much less independent information than 8,760 independent draws would. Methods that assume independence then understate their own uncertainty: standard errors come out too small and results look more significant than they are. Chapter 12 treats this in regression and Chapter 15 models it directly. Cross-sectional units can be dependent too, for example wind farms in the same weather system, so independence should be checked rather than assumed; but the time ordering makes dependence the default for (a) and (c).

Solution 1.5 (Survivorship bias)**.**

(a) Funds that performed badly were disproportionately closed or merged away during the decade, and they have dropped out of the database. The survivors are therefore selected partly on their returns, and their average return sits above that of the full set of funds that existed ten years ago. The bias is upward.

(b) No. The additional funds were filtered by the same survival mechanism, so the enlarged sample estimates the same wrong quantity with more precision. A larger biased sample gives a more precise estimate of the wrong number.

(c) The returns of the funds that died, up to the date they closed. In practice this means a "survivorship-bias-free" database that keeps defunct funds. The population of interest (all funds alive ten years ago) must be the population from which the sample is drawn.

Solution 1.6 (Who answered the survey?)**.**

(a) Three sources stand out. *Selection through the sampling frame:* only customers who called the call center are surveyed, and people mostly call when something is wrong. *Nonresponse:* only those who choose to reply are counted, and the decision to reply is unlikely to be unrelated to how the customer feels. *Coverage:* customers without an email address on file cannot be reached at all.

(b) The frame effect almost certainly pushes the estimate down, because callers are disproportionately customers with problems. Nonresponse usually also pushes it down, since aggrieved customers are more motivated to reply, but its direction is not guaranteed; that is precisely why it is dangerous. The coverage effect has no obvious sign; it depends on whether customers without email (often older or less engaged) are more or less satisfied. The overall conclusion is that 18% is very likely an underestimate and, more importantly, that its error cannot be computed from the data; it can only be reduced by drawing a random sample of all customers and following up with non-respondents.

Solution 1.7 (Random error versus bias)**.**

(a) Meter A. Its errors are random with mean zero, so they cancel on averaging. Meter B's errors all have the same sign and size, so averaging leaves them untouched.

(b) Meter A's average has typical error $2/\sqrt{10{,}000} = 0.02$ units. Meter B's average is still exactly 1.5 units too high. Meter A wins by a factor of 75, even though any single reading from B looks more precise.

(c) Only random (sampling) error. A confidence interval built from Meter B's readings would be very narrow and centered on the wrong value; nothing in the data reveals the bias. Detecting it requires an outside check such as calibration against a known standard.

Solution 1.8 (Two samples, two answers)**.**

(a) Not necessarily. Both averages are correct descriptions of their own samples. Neither equals μ, and neither analyst should claim it does.

(b) Sampling variability: a statistic computed from a sample changes when the sample changes. The two values are two draws from the sampling distribution of $\bar{X}$.

(c) Quadrupling n from 30 to 120 multiplies $1/\sqrt{n}$ by $1/2$, so the typical sampling error of each average halves, and so does the typical gap between the two averages. A gap of about 0.35 would be typical rather than 0.7. This holds only if the observations are roughly independent; strongly autocorrelated daily data shrink the error more slowly.

Solution 1.9 (Bias does not wash out)**.**

(a) For $n = 25$: $\sqrt{9 + (10/5)^2} = \sqrt{13} \approx 3.61$. For $n = 100$: $\sqrt{9+1} = \sqrt{10} \approx 3.16$. For $n = 2{,}500$: $\sqrt{9 + 0.04} \approx 3.01$.

(b) It approaches 3, the size of the bias. A hundredfold increase in sample size (25 to 2,500) improves the error by only 17%. Beyond a modest n, more data from the same biased procedure buys almost nothing; the only remedy is to fix the procedure.

Solution 1.10 (Defining the population)**.**

(a) The population is the set of daily spreads that the current market process can generate, in particular over the future period for which the answer will be used. The sample is the roughly 1,800 observed daily spreads. The parameter is a population measure of spread such as the true standard deviation σ of the daily spread (or of its daily change, if that is what "volatile" means; the question should be settled before any computation).

(b) The values of most interest, future spreads, do not yet exist. The population is a model of what the data-generating process could produce, not a list of existing units that could in principle be enumerated.

(c) Treating five years of history as a random sample of the future assumes the process is stable, and several things can break that. Market rules, transmission capacity, or the fuel mix can change, shifting the level and the volatility of the spread. Regimes come and go: a single extreme winter can dominate five years of data, or be absent from them. And consecutive days are dependent, so 1,800 observations carry less information than 1,800 independent draws. Chapter 15 (stationarity and time series) addresses the stability question directly, and Chapter 12 addresses the effect of dependence on standard errors.

Chapter 2

Describing Data

Solution 2.1 (Center by hand)**.** The sum is 44, so $\bar{x} = 44/8 = 5.5$. Sorted, the data are $2, 3, 4, 5, 7, 7, 7, 9$; with $n = 8$ the median is the average of the 4th and 5th values, $(5 + 7)/2 = 6$. The mode is 7, which occurs three times.

Here mean $<$ median $<$ mode. A mean below the median points to a longer left tail: the low values $2, 3, 4$ pull the mean down, while the upper values are bunched at 7. With eight observations this is a weak signal, not a firm conclusion about the population.

Solution 2.2 (Spread by hand)**.** With $\bar{x} = 5.5$ the deviations, in the original order, are

$$-2.5,\ 1.5,\ 1.5,\ -3.5,\ 3.5,\ -1.5,\ 1.5,\ -0.5,$$

which sum to zero as they must. Their squares are $6.25, 2.25, 2.25, 12.25, 12.25, 2.25, 2.25, 0.25$, summing to 40. Hence

$$s^2 = \frac{40}{8-1} = 5.714, \qquad s = \sqrt{5.714} = 2.390,$$

and dividing by n gives $40/8 = 5.0$.

Check: $\sum x_i^2 = 9 + 49 + 49 + 4 + 81 + 16 + 49 + 25 = 282$ and $n\bar{x}^2 = 8 \times 30.25 = 242$, so $\sum (x_i - \bar{x})^2 = 282 - 242 = 40$, as before.

Solution 2.3 (One bad value)**.** *Original data.* Sorted: $2, 3, 4, 5 \mid 7, 7, 7, 9$. The lower half has median $Q_1 = (3 + 4)/2 = 3.5$ and the upper half $Q_3 = (7 + 7)/2 = 7$, so IQR $= 3.5$. The absolute deviations from the median 6 are $3, 1, 1, 4, 3, 2, 1, 1$; sorted, $1, 1, 1, 1, 2, 3, 3, 4$, with median MAD $= 1.5$.

Contaminated data. The sum rises by 81 to 125, so the mean becomes $125/8 = 15.625$. The median is still 6, and Q_1, Q_3, IQR and MAD are unchanged, because the halves and the ranks of the middle values do not change. The sum of squared deviations becomes 6,347.9, so $s^2 = 906.8$ and $s = 30.11$.

	original	with 90
mean	5.5	15.625
median	6	6
s	2.39	30.11
IQR	3.5	3.5
MAD	1.5	1.5

One bad value nearly triples the mean and multiplies the standard deviation by more than twelve, while the robust summaries do not move at all. The fences are $Q_1 - 1.5\,\text{IQR} = -1.75$ and $Q_3 + 1.5\,\text{IQR} = 12.25$, so 90 is flagged, which is exactly what the rule is for.

Solution 2.4 (Outage durations)**.**

(a) Sorted: $0.5, 0.8, 0.9, 1.1, 1.2, 1.4 \mid 1.7, 1.9, 2.2, 2.5, 3.1, 9.6$. The median is $(1.4 + 1.7)/2 = 1.55$. The lower half has median $Q_1 = (0.9 + 1.1)/2 = 1.0$ and the upper half $Q_3 = (2.2 + 2.5)/2 = 2.35$, so IQR $= 1.35$.

(b) The fences are $1.0 - 1.5(1.35) = -1.025$ and $2.35 + 1.5(1.35) = 4.375$. Only the 9.6-hour outage lies outside them.

(c) The sum is 26.9, so $\bar{x} = 2.24$, well above the median of 1.55. The long right tail, driven by the 9.6, pulls the mean up: the distribution is right-skewed, as durations usually are.

(d) Report the median, about 1.5 hours, as the typical outage, because it describes the usual case and is not dragged by a single extreme event. Report the 9.6-hour event separately, not discard it: for planning and risk it may be the most important observation in the set.

Solution 2.5 (Volume-weighted price)**.** The weighted mean is

$$\bar{x}_w = \frac{100(40.00) + 300(40.50) + 600(41.00)}{100 + 300 + 600} = \frac{40{,}750}{1{,}000} = \$40.75,$$

while the simple average of the three prices is $(40.00 + 40.50 + 41.00)/3 = \$40.50$. The volume-weighted figure is the true average cost per MWh: total dollars paid divided by total MWh bought. The simple average treats a 100 MWh fill as if it mattered as much as a 600 MWh fill. Because the largest volume traded at the highest price, the weighted average is pulled up toward \$41.

Solution 2.6 (Two characterizations of the mean)**.**

(a) $\sum_i (x_i - c) = \sum_i x_i - nc = n\bar{x} - nc = n(\bar{x} - c)$, which is zero exactly when $c = \bar{x}$.

(b) Write $x_i - c = (x_i - \bar{x}) + (\bar{x} - c)$ and expand:

$$\sum_i (x_i - c)^2 = \sum_i (x_i - \bar{x})^2 + 2(\bar{x} - c)\sum_i (x_i - \bar{x}) + n(\bar{x} - c)^2.$$

The middle sum is zero by (a). The first term does not involve c and the last is non-negative, equal to zero only when $c = \bar{x}$; hence $c = \bar{x}$ is the unique minimizer.

(c) With $c = 0$: $\sum x_i^2 = \sum (x_i - \bar{x})^2 + n\bar{x}^2$, that is, $\sum (x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2$.

The decomposition in (b) is the same "add and subtract the mean" move that produces the bias–variance decomposition in Chapter 7 and the sum-of-squares partition in Chapter 10.

Solution 2.7 (The median minimizes absolute deviations)**.**

(a) $S(3) = 2 + 1 + 0 + 1 + 97 = 101$; $S(4) = 3 + 2 + 1 + 0 + 96 = 102$; $S(22) = 21 + 20 + 19 + 18 + 78 = 156$.

(b) Move c to the right by a small amount δ, staying between data points. Every observation to the left of c has its distance increased by δ, and every observation to the right has its distance decreased by δ. So S changes by $\delta \times (\#\text{left} - \#\text{right})$. To the right of the median at least half the points lie to the left, so S cannot decrease; by symmetry the same holds for moves to the left. Hence S is minimized at the median. Only the *count* of points on each side matters, not how far away they are, which is why the outlier at 100 has no pull on the median.

Solution 2.8 (Changing units)**.**

(a) The mean transforms like the data: $1.8(20) + 32 = 68°\text{F}$. Adding a constant does not change spread, and multiplying by 1.8 multiplies the standard deviation by 1.8: $s = 9°\text{F}$, so the variance is 81 (in $°\text{F}^2$), versus 25 (in $°\text{C}^2$).

(b) $\text{CV}_C = 5/20 = 0.25$ but $\text{CV}_F = 9/68 = 0.13$. The CV divides by the mean, and the mean depends on where zero is placed; both scales have an arbitrary zero, so "standard deviation as a fraction of the mean" has no physical meaning. The CV is meaningful only on a ratio scale with a true zero, such as prices, volumes, or kelvins.

(c) In Celsius, $z = (30 - 20)/5 = 2$. The same day is 86°F, and $z = (86 - 68)/9 = 2$. A z-score subtracts the mean and divides by the standard deviation, which removes both the shift and the scale of any linear change of units.

Solution 2.9 (Which is more unusual?)**.**

(a) $z_{\text{plant}} = (820 - 700)/80 = 1.5$ and $z_{\text{price}} = (62 - 50)/4 = 3.0$. The price is twice as many standard deviations from its mean, so it is the more unusual observation, even though the plant's deviation of 120 MWh is numerically much larger than \$12.

(b) A z-score converts to a tail probability only through a distributional assumption, typically normality (Chapter 5). Under a normal model, $z = 3$ is about a 1-in-740 upper-tail event. If prices are skewed or heavy-tailed, as they often are, a z of 3 can be far more common than the normal suggests, so the ranking in (a) is reliable but the probability statement is not.

Solution 2.10 (Skewness and kurtosis)**.** *All ten values.* $\bar{x} = 50/10 = 5$ and the median is 4. The deviations from 5 are $-3, -2, -2, -1, -1, -1, 0, 0, 1, 9$. Their squares sum to 102, so $s^2 = 102/9 = 11.33$ and $s = 3.367$. Their cubes sum to 684, so

$$g_1 = \frac{1}{10} \cdot \frac{684}{s^3} = \frac{68.4}{38.16} = 1.79.$$

Their fourth powers sum to 6,678, so $g_2 = 667.8/s^4 = 667.8/128.4 = 5.20$ and the excess kurtosis is 2.20.

Without the 14. The nine values have mean 4, median 4, $s = 1.225$, $g_1 = 0$ (the remaining data are exactly symmetric about 4), and $g_2 = 1.78$, so the excess kurtosis is -1.22.

A single observation turns a symmetric, light-tailed sample into one with strong positive skew and heavy tails. Because they use cubes and fourth powers of deviations, g_1 and g_2 are even more sensitive to extreme values than s, and in small samples they are dominated by the one or two most extreme points.

Solution 2.11 (Reading the ECDF)**.** Sorted data: $2, 3, 4, 5, 7, 7, 7, 9$, with $n = 8$. Four observations are ≤ 5, so $\hat{F}(5) = 4/8 = 0.5$. No observation lies in $(5, 6.5]$, so $\hat{F}(6.5) = 0.5$ as well; the ECDF is flat between data points. Seven observations are ≤ 7, so $\hat{F}(7) = 0.875$. Since $\hat{F}$ is 0.5 just below 7 and 0.875 at 7, the smallest x with $\hat{F}(x) \geq 0.75$ is $x = 7$: the 0.75 quantile read off the ECDF is 7. The jump at 7 is $3/8$, because three tied observations sit there; in general each jump is $1/n$ times the number of observations at that value.

Solution 2.12 (Reading shapes)**.**

(a) Mean well above median: right skew, with a long upper tail of large values.

(b) The extreme sample quantiles are more extreme than the normal predicts in both directions: heavy tails (positive excess kurtosis), roughly symmetric.

(c) The lower quantiles are less extreme than normal (points above the line on the left) and the upper quantiles more extreme (above the line on the right): right skew. Strictly, skewness bends the plot into a single convex or concave curve, while heavy or light tails produce the S-shape of (b); the two signatures are easy to confuse, and some texts, including the one-line description in Section 2.4.4, use "S-shape" loosely for skewness.

(d) Right skew: the middle half is bunched near its lower end, and the upper tail stretches far.

(e) Bimodality, which often means a mixture of two different groups (for example peak and off-peak hours) that should be analyzed separately. A boxplot, built only from quartiles, cannot show it: it would display a single box, possibly with a wide IQR, and the two peaks would be invisible.

Solution 2.13 (Same mean and variance, different shape)**.**

(a) D_1 sums to 0 and its squares sum to 4, so $s^2 = 4/3$. For D_2, $-\sqrt{3} + 3/\sqrt{3} = -\sqrt{3} + \sqrt{3} = 0$, and the squares sum to $3 + 3 \cdot \frac{1}{3} = 4$, so again $s^2 = 4/3$.

(b) For both, $s = \sqrt{4/3}$ and $s^3 = (4/3)^{3/2} = 1.540$. For D_1 the cubes are $-1, -1, 1, 1$, summing to 0, so $g_1 = 0$. For D_2 the cubes sum to $-3\sqrt{3} + 3/(3\sqrt{3}) = -3\sqrt{3} + 1/\sqrt{3} = -4.619$, so $g_1 = \frac{1}{4}(-4.619)/1.540 = -0.75$.

(c) Matching the first two moments does not pin down a distribution. D_1 is symmetric; D_2 has one large negative value balanced by several small positive ones, a left-skewed shape typical of, for example, the returns of a strategy that sells insurance. Center and spread alone would report the two as identical. This is the reason to compute shape statistics and, above all, to plot the data.

Chapter 3

Probability

Solution 3.1 (Two dice)**.** The sample space has 36 equally likely ordered pairs.

(a) Six pairs sum to 7: $(1,6), (2,5), \ldots, (6,1)$. $P = 6/36 = 1/6$.

(b) By the complement rule, $P(\text{no six}) = (5/6)^2 = 25/36$, so $P(\text{at least one six}) = 11/36$.

(c) Sums of $10, 11, 12$ arise from $3 + 2 + 1 = 6$ pairs: $P = 6/36 = 1/6$.

(d) Restrict to the 11 pairs containing a six. Of these, $(4,6), (6,4), (5,6), (6,5), (6,6)$ have sum ≥ 10, so $P = 5/11 \approx 0.455$.

(e) No. Conditioning on a six raises the probability of a high sum from $1/6 \approx 0.167$ to $5/11 \approx 0.455$. Equivalently, $P(\text{both}) = 5/36 \neq (1/6)(11/36)$.

Solution 3.2 (Inclusion–exclusion)**.**

(a) From $P(A \cup B) = P(A) + P(B) - P(A \cap B)$: $P(A \cap B) = 0.3 + 0.4 - 0.55 = 0.15$.

(b) $1 - P(A \cup B) = 0.45$.

(c) $P(A \cup B) - P(A \cap B) = 0.55 - 0.15 = 0.40$.

(d) $P(A \mid B) = 0.15/0.4 = 0.375$ and $P(B \mid A) = 0.15/0.3 = 0.5$. Note that the two conditionals differ.

(e) Independence would require $P(A \cap B) = P(A)P(B) = 0.12$. The actual 0.15 is larger, so they are dependent and positively related: learning B raises the probability of A from 0.30 to 0.375.

Solution 3.3 (At least one outage)**.**

(a) The complement is "no outage in any month," with probability $0.9^{12} = 0.282$ by independence. So $P(\text{at least one}) = 1 - 0.282 = 0.718$.

(b) 1.2 is the *expected number* of outages in a year (linearity of expectation, Chapter 4), a perfectly good quantity but not a probability. Adding $P(A_1) + \cdots + P(A_{12})$ treats the monthly events as mutually exclusive, which double-counts years with several outages. The sum is only an upper bound on $P(\text{at least one})$, the union bound of Exercise 3.13.

Solution 3.4 (Counting portfolios)**.**

(a) Order does not matter: $\binom{10}{3} = 120$.

(b) Order matters: $10 \cdot 9 \cdot 8 = 10!/7! = 720$, which is $3! = 6$ times the answer to (a), because each portfolio can be ranked in 6 ways.

(c) If both particular assets are in, the third member is any of the other 8: $8/120 = 1/15 \approx 0.067$. If neither is in, all three come from the other 8: $\binom{8}{3}/120 = 56/120 = 7/15 \approx 0.467$.

Solution 3.5 (Clashing choices)**.** There are 10^5 equally likely assignments. The number with all days different is $10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 = 30{,}240$, so

$$P(\text{all different}) = \frac{30{,}240}{100{,}000} = 0.302, \qquad P(\text{some clash}) = 1 - 0.302 = 0.698.$$

A clash is more likely than not with only five people and ten days. This is the logic of the "birthday problem": the number of *pairs* that could coincide, $\binom{5}{2} = 10$, grows much faster than the number of people.

Solution 3.6 (Conditional probabilities from a table)**.** Using relative frequencies: $P(\text{spike}) = 27/250 = 0.108$; $P(\text{spike} \mid \text{high}) = 18/60 = 0.30$; $P(\text{spike} \mid \text{calm}) = 9/190 = 0.047$; $P(\text{high} \mid \text{spike}) = 18/27 = 0.667$.

Not independent: the spike rate is more than six times higher on high-volatility days than on calm days, and both differ from the overall 0.108. (Chapter 8, Exercise 8.10, tests whether a difference this large could be chance.)

The two conditionals answer different questions. Only 30% of high-volatility days have spikes, but two-thirds of spikes occur on high-volatility days. The second number is large partly because spikes are rare on calm days, and calm days, though more common, contribute few spikes. Confusing the two is the inversion fallacy of Section 3.6.

Solution 3.7 (Disjoint is not independent)**.** Mutual exclusivity gives $P(A \cap B) = P(\varnothing) = 0$, while $P(A)P(B) > 0$ because both factors are positive. So $P(A \cap B) \neq P(A)P(B)$. Intuitively, learning that A occurred tells you for certain that B did not, which is the opposite of carrying no information: $P(B \mid A) = 0 < P(B)$.

Solution 3.8 (Regimes and the law of total probability)**.**

(a) $P(S) = P(S \mid H)P(H) + P(S \mid C)P(C) = 0.25(0.2) + 0.02(0.8) = 0.050 + 0.016 = 0.066$.

(b) By Bayes' theorem, $P(H \mid S) = 0.050/0.066 = 0.758$.

(c) $P(H \mid S^c) = \dfrac{P(S^c \mid H)P(H)}{1 - P(S)} = \dfrac{0.75(0.2)}{0.934} = 0.161$.

Seeing a spike moves the probability of the high-volatility regime from the prior 0.20 to 0.76; seeing no spike moves it only slightly, to 0.16, because quiet days are common in both regimes. Rare events are informative when they occur; their absence is weak evidence.

Solution 3.9 (Testing twice)**.**

(a) $P(+) = 0.99(0.001) + 0.01(0.999) = 0.00099 + 0.00999 = 0.01098$, so

$$P(\text{present} \mid +) = \frac{0.00099}{0.01098} = 0.090.$$

(b) Expect $100{,}000 \times 0.001 \times 0.99 = 99$ true positives and $100{,}000 \times 0.999 \times 0.01 = 999$ false positives. About one positive in eleven is real, which is the answer to (a) seen as counts.

(c) Use 0.090 as the new prior:

$$P(\text{present} \mid ++) = \frac{0.99(0.0902)}{0.99(0.0902) + 0.01(0.9098)} = \frac{0.0893}{0.0893 + 0.0091} = 0.908.$$

A second positive raises the probability from 9% to 91%.

(d) Whatever caused a false positive the first time (a cross-reacting substance, a quirk of the person) is likely to cause it again. Positive dependence between the errors makes the second positive less informative than assumed, so 0.908 is an overestimate. A second test of a *different* kind is worth more than a repeat of the same one.

Solution 3.10 (The inversion fallacy)**.**

(a) It confuses $P(\text{high load} \mid \text{spike}) = 0.95$ with $P(\text{spike} \mid \text{high load})$. The two are linked by Bayes' theorem through the base rates $P(\text{spike})$ and $P(\text{high load})$, and they can be very different.

(b) $P(\text{spike} \mid \text{high}) = \dfrac{P(\text{high} \mid \text{spike})P(\text{spike})}{P(\text{high})} = \dfrac{0.95(0.02)}{0.25} = 0.076.$

High load is nearly necessary for a spike, but far from sufficient: fewer than 1 high-load day in 13 has a spike. A trading rule built on the 95% figure would be wrong most of the time.

Solution 3.11 (Pairwise is not mutual independence)**.** The four outcomes HH, HT, TH, TT are equally likely.

(a) $P(A) = P(B) = P(C) = 1/2$. $A \cap B = \{HH\}$, $A \cap C = \{HT\}$, $B \cap C = \{TH\}$, each with probability $1/4 = (1/2)(1/2)$. So every pair is independent.

(b) $A \cap B \cap C = \varnothing$ (two heads is not exactly one head), so $P(A \cap B \cap C) = 0 \neq 1/8$.

(c) Any one event tells you nothing about any other single event, but any two together determine the third completely. Independence of a collection is a statement about all sub-collections, not just pairs, which matters whenever probabilities of many joint events are multiplied together.

Solution 3.12 (Three doors)**.** Let D_k be "prize behind door k" (prior $1/3$ each) and O_3 "host opens door 3." The likelihoods are $P(O_3 \mid D_1) = 1/2$ (he may open 2 or 3), $P(O_3 \mid D_2) = 1$ (door 3 is his only option), and $P(O_3 \mid D_3) = 0$. By the law of total probability $P(O_3) = \frac{1}{3}(\frac{1}{2} + 1 + 0) = \frac{1}{2}$, so

$$P(D_2 \mid O_3) = \frac{1 \cdot \frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}, \qquad P(D_1 \mid O_3) = \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{2}} = \frac{1}{3}.$$

Switch: it wins with probability $2/3$. The host's action is evidence because it depends on where the prize is; it carries no information about your door, which keeps its prior $1/3$, but it concentrates the remaining $2/3$ on door 2. A simulation of 200,000 games gives a switching win rate of 0.665.

Solution 3.13 (Two bounds from the axioms)**.**

(a) By inclusion–exclusion, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, and $P(A \cap B) \geq 0$ by non-negativity.

(b) Rearranging inclusion–exclusion, $P(A \cap B) = P(A) + P(B) - P(A \cup B) \geq P(A) + P(B) - 1$, since $P(A \cup B) \leq 1$.

(c) By (b), $P(\text{both correct}) \geq 0.9 + 0.9 - 1 = 0.8$, whatever the dependence. The upper bound is 0.9 (both events always coincide), and independence would give 0.81. Part (a) is the union bound behind the Bonferroni correction of Chapters 8 and 13: the chance of at least one false positive among m tests is at most the sum of the individual error rates.

Chapter 4

Random Variables

Solution 4.1 (A discrete distribution)**.**

(a) The probabilities must sum to one: $c = 1 - 0.8 = 0.2$.

(b) $\mathbb{E}[X] = 0(0.1) + 1(0.3) + 2(0.4) + 3(0.2) = 1.7$. $\mathbb{E}[X^2] = 0 + 0.3 + 1.6 + 1.8 = 3.7$. $\mathrm{Var}(X) = 3.7 - 1.7^2 = 0.81$, so $\mathrm{SD}(X) = 0.9$.

(c) $F(1.5) = P(X \le 1.5) = P(X = 0) + P(X = 1) = 0.4$, and $P(X > 1) = 1 - F(1) = 0.6$.

(d) $\mathbb{E}[1/(1+X)] = 0.1/1 + 0.3/2 + 0.4/3 + 0.2/4 = 0.433$, whereas $1/(1+1.7) = 0.370$. The function $g(x) = 1/(1+x)$ is convex for $x \ge 0$, so by Jensen's inequality $\mathbb{E}[g(X)] \ge g(\mathbb{E}[X])$. The expectation of a nonlinear function must be computed from the distribution, not by plugging in the mean.

Solution 4.2 (A continuous distribution)**.**

(a) $\int_0^2 cx^2\,dx = 8c/3 = 1$ gives $c = 3/8$. For $0 \le x \le 2$, $F(x) = \int_0^x \frac{3}{8}t^2\,dt = x^3/8$; $F(x) = 0$ for $x < 0$ and 1 for $x > 2$.

(b) $F(2) - F(1) = 1 - 1/8 = 7/8$.

(c) $\mathbb{E}[X] = \int_0^2 x \cdot \frac{3}{8}x^2\,dx = \frac{3}{8} \cdot \frac{16}{4} = 1.5$. $\mathbb{E}[X^2] = \frac{3}{8} \cdot \frac{32}{5} = 2.4$. $\mathrm{Var}(X) = 2.4 - 2.25 = 0.15$.

(d) Solve $x^3/8 = 1/2$: $x = 4^{1/3} \approx 1.587$. The median exceeds the mean 1.5 because the density has a long lower tail: it is left-skewed.

(e) $f(2) = 3/8 \cdot 4 = 1.5$. No problem: a density is probability per unit length, not a probability. Probabilities come from areas, and the total area is one.

Solution 4.3 (Linear transformations)**.**

(a) $\mathbb{E}[Y] = 3 - 4(10) = -37$. $\mathrm{Var}(Y) = (-4)^2\,\mathrm{Var}(X) = 16 \times 4 = 64$, so $\mathrm{SD}(Y) = 8 = |-4| \times 2$. The sign of the multiplier reverses the direction but cannot make the spread negative.

(b) $\mathbb{E}[\text{profit}] = 5{,}000(40 - 38) - 2{,}000 = \$8{,}000$. The constants -38 and $-2{,}000$ shift the distribution without changing its spread, so $\mathrm{SD}(\text{profit}) = 5{,}000 \times 6 = \$30{,}000$. The position is expected to make money, but a one-standard-deviation move against it costs nearly four times the expected profit.

Solution 4.4 (Jensen's inequality and compounding)**.**

(a) $\mathbb{E}[M] = \frac{1}{2}(0.5) + \frac{1}{2}(1.5) = 1$. Wealth after n periods is the product $M_1 \cdots M_n$ of independent multipliers, and the expectation of a product of independent variables is the product of expectations, so expected wealth is $1^n = 1$ for every n.

(b) $\mathbb{E}[\ln M] = \frac{1}{2}(\ln 0.5 + \ln 1.5) = \frac{1}{2}(-0.693 + 0.405) = -0.144$, while $\ln \mathbb{E}[M] = \ln 1 = 0$. Since ln is concave, Jensen's inequality gives $\mathbb{E}[\ln M] \le \ln \mathbb{E}[M]$; here the gap is large. The per-period growth factor of a typical path is $e^{-0.144} = 0.866 = \sqrt{0.5 \times 1.5}$, the geometric mean of the two multipliers.

(c) The median path has 50 up-moves and 50 down-moves: $1.5^{50} \times 0.5^{50} = 0.75^{50} \approx 5.7 \times 10^{-7}$.

(d) The mean is held at 1 by a handful of astronomically lucky paths; the typical path is wiped out. Wealth exceeds its starting value only with 64 or more up-moves out of 100, which has probability about 0.003 (a binomial calculation, Chapter 5). Expected value is the wrong criterion for a bet you must take repeatedly with compounding stakes; the expected log return, which is negative, governs what happens on almost every path. This is the arithmetic-versus-geometric gap mentioned in Section 4.2.

Solution 4.5 (A joint distribution)**.**

(a) Sum across rows and columns: $P(X=1) = 0.2 + 0.3 = 0.5$ and $P(Y=1) = 0.1 + 0.3 = 0.4$.

(b) $P(Y=1 \mid X=1) = 0.3/0.5 = 0.6$ and $P(Y=1 \mid X=0) = 0.1/0.5 = 0.2$.

(c) No: the conditional distribution of Y depends on X. Equivalently, $p(1,1) = 0.3 \neq P(X=1)P(Y=1) = 0.2$.

(d) For indicators, $\mathbb{E}[XY] = P(X=1, Y=1) = 0.3$, so $\mathrm{Cov}(X,Y) = 0.3 - 0.5(0.4) = 0.1$. The variances are $0.5(0.5) = 0.25$ and $0.4(0.6) = 0.24$, so $\mathrm{Corr}(X,Y) = 0.1/\sqrt{0.25 \times 0.24} = 0.408$.

Solution 4.6 (Variance of a sum)**.** $\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \pm 2\,\mathrm{Cov}(X,Y)$.

Cov	$\mathrm{Var}(X+Y)$	$\mathrm{SD}(X+Y)$	$\mathrm{Var}(X-Y)$
-3	7	2.65	19
0	13	3.61	13
$+3$	19	4.36	7

Standard deviations do not add: even with positive covariance, $\mathrm{SD}(X+Y) = 4.36 < 2 + 3$. Combining positions that move oppositely (negative covariance, or a short position in something positively correlated) reduces risk. That is the logic of a hedge, developed further in Exercise 9.5.

Solution 4.7 (Diversification has a floor)**.**

(a) The portfolio return is $\frac{1}{n}\sum_i R_i$. Its variance is $\frac{1}{n^2}\left[\sum_i \mathrm{Var}(R_i) + \sum_{i \neq j} \mathrm{Cov}(R_i, R_j)\right] = \frac{1}{n^2}\left[n\sigma^2 + n(n-1)\rho\sigma^2\right]$, which simplifies to the stated expression.

(b) The variance is $0.04/n + (1 - 1/n)(0.012)$: $n = 1$: 20.0%; $n = 10$: $\sqrt{0.0148} = 12.2\%$; $n = 100$: $\sqrt{0.01228} = 11.1\%$; limit: $\sqrt{\rho}\,\sigma = \sqrt{0.012} = 11.0\%$.

The first term, idiosyncratic risk, vanishes like $1/n$; the second, common risk, does not diversify away. Most of the benefit arrives with the first ten or so assets. With $\rho = 0$ the formula reduces to the σ^2/n of Section 4.5; dependence is what puts a floor under the risk.

Solution 4.8 (Standard error of an average)**.**

(a) $\sigma/\sqrt{n} = 2\%/5 = 0.4\%$.

(b) Solve $2\%/\sqrt{n} = 0.1\%$: $\sqrt{n} = 20$, $n = 400$ days.

(c) It halves. Going from a standard error of 0.4% to 0.1% requires sixteen times as much data. Note that 400 trading days is well over a year: a mean daily return is very hard to estimate precisely, even when the volatility is easy to estimate. (If the returns were autocorrelated, even more data would be needed.)

Solution 4.9 (Moment generating functions)**.**

(a) $M(t) = \mathbb{E}[e^{tX}] = (1-p)e^0 + pe^t = 1 - p + pe^t$. Then $M'(t) = pe^t$ and $M''(t) = pe^t$, so $\mathbb{E}[X] = M'(0) = p$ and $\mathbb{E}[X^2] = M''(0) = p$, giving $\mathrm{Var}(X) = p - p^2 = p(1-p)$.

(b) MGFs of independent variables multiply: $M_S(t) = (1 - p + pe^t)^n$. This is the binomial MGF, so the sum is Binomial(n, p); uniqueness of MGFs makes the identification rigorous.

(c) $M_{X+Y}(t) = M_X(t)M_Y(t) = \exp\left((\mu_1 + \mu_2)t + (\sigma_1^2 + \sigma_2^2)t^2/2\right)$, which is the MGF of $N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$. By uniqueness, $X + Y$ has that distribution.

Solution 4.10 (Linearity without independence)**.** Let $I_k = 1$ if letter k lands in its own envelope. By symmetry $P(I_k = 1) = 1/n$, so $\mathbb{E}[I_k] = 1/n$. Since $N = \sum_{k=1}^{n} I_k$, linearity gives $\mathbb{E}[N] = n \cdot \frac{1}{n} = 1$.

The indicators are dependent: if the first $n - 1$ letters are all correct, the last must be too. Linearity of expectation does not care; it holds for any random variables. (A check by enumeration: for $n = 3$ the six arrangements have $3, 1, 1, 1, 0, 0$ correct letters, an average of exactly 1.) The variance of N, by contrast, does depend on the covariances between the I_k.

Solution 4.11 (Uncorrelated but dependent)**.**

(a) $\mathbb{E}[X] = 0$ and $\mathbb{E}[XY] = \mathbb{E}[X^3] = \frac{1}{3}(-1 + 0 + 1) = 0$, so $\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0$.

(b) $P(Y = 1) = 2/3$ but $P(Y = 1 \mid X = 0) = 0$. Knowing X determines Y completely.

Independence implies zero covariance, but not conversely. Covariance measures only linear co-movement, a point Chapter 9 makes about correlation.

Solution 4.12 (From CDF to PMF)**.** The PMF is the size of each jump: $p(0) = 0.20$, $p(1) = 0.30$, $p(2) = 0.35$, $p(3) = 0.15$. Then $\mathbb{E}[X] = 0.30 + 0.70 + 0.45 = 1.45$. $P(1 < X \leq 3) = F(3) - F(1) = 0.50$, and $P(X \geq 2) = 1 - F(1) = 0.50$. With a discrete variable, take care with strict versus weak inequalities: $P(X \geq 2) = 1 - F(1)$, not $1 - F(2)$.

Solution 4.13 (A threshold probability)**.**

(a) $P(X > 60) = 1 - F(60) = (30/60)^3 = 1/8 = 0.125$ and $P(X > 90) = (1/3)^3 = 1/27 = 0.037$.

(b) Solve $(30/x)^3 = 1/2$: $x = 30 \cdot 2^{1/3} = \$37.80$.

(c) $f(x) = F'(x) = 3 \cdot 30^3/x^4$ for $x \geq 30$. Then $\mathbb{E}[X] = \int_{30}^{\infty} x \cdot 3 \cdot 30^3 x^{-4}\, dx = 3 \cdot 30^3 \left[\frac{x^{-2}}{-2}\right]_{30}^{\infty} = \frac{3}{2} \cdot 30 = \$45$.

The mean (\$45) is well above the median (\$37.80): this Pareto-type distribution is strongly right-skewed. Its tail decays only like x^{-3}, far more slowly than a normal tail, so a doubling of price is a 1-in-8 event rather than a negligible one. With an exponent of 2 instead of 3 the variance would be infinite, the situation in which the Central Limit Theorem of Chapter 6 fails.

Chapter 5

The Distributions That Matter

Solution 5.1 (A binary contract)**.**

(a) The payoff is $100X$ with $X \sim \text{Bernoulli}(0.3)$. Mean $100(0.3) = \$30$; standard deviation $100\sqrt{0.3 \times 0.7} = \$45.83$.

(b) $p(1-p)$ is maximized at $p = 1/2$, giving $100 \times 0.5 = \$50$.

(c) It rises to the peak of \$50 at $p = 0.5$ and then falls, to $100\sqrt{0.9 \times 0.1} = \$30$ at $p = 0.9$. Uncertainty about a binary outcome is greatest at a coin flip and shrinks as the outcome becomes nearly certain in either direction.

Solution 5.2 (Counting failures)**.** Let $X \sim \text{Binomial}(10, 0.1)$.

(a) $P(X = 0) = 0.9^{10} = 0.349$. $P(X = 2) = \binom{10}{2}(0.1)^2(0.9)^8 = 45(0.01)(0.4305) = 0.194$. $P(X = 1) = 10(0.1)(0.9)^9 = 0.387$, so $P(X \geq 2) = 1 - 0.349 - 0.387 = 0.264$.

(b) $\mathbb{E}[X] = np = 1$ and $\text{Var}(X) = np(1-p) = 0.9$.

(c) Independence. Units at one site share weather, maintenance practices, fuel supply, and design flaws, so failures tend to cluster (common-mode failure). Positive dependence leaves the mean at 1 but increases the variance and makes both zero failures and many failures more likely than the binomial says. Risk calculations that assume independence understate the chance of multiple simultaneous failures.

Solution 5.3 (Waiting for success)**.** Let $X \sim \text{Geometric}(0.2)$.

(a) $\mathbb{E}[X] = 1/p = 5$; $\text{Var}(X) = (1-p)/p^2 = 0.8/0.04 = 20$, so $\text{SD}(X) = 4.47$. The spread is almost as large as the mean: waiting times are highly variable.

(b) Two failures then a success: $0.8^2(0.2) = 0.128$.

(c) More than five attempts means the first five all fail: $0.8^5 = 0.328$. One time in three you wait more than five attempts, although the average is five.

Solution 5.4 (Poisson counts)**.**

(a) With $X \sim \text{Poisson}(2)$: $P(X = 0) = e^{-2} = 0.135$. $P(X \leq 2) = e^{-2}(1 + 2 + 2) = 5e^{-2} = 0.677$, so $P(X \geq 3) = 0.323$.

(b) Over a quarter the count is Poisson with mean $3 \times 2 = 6$, so $P(0) = e^{-6} = 0.0025$, about once in 400 quarters.

(c) Each waiting time is Exponential(2) with mean $1/2$ month and variance $1/4$. The sum of three independent ones is Gamma with shape 3 and scale $1/2$: mean $3/2 = 1.5$ months, variance $3/4$, standard deviation 0.87 months.

Solution 5.5 (The Poisson approximation)**.** Here $\lambda = np = 2$.

k	Binomial(500, 0.004)	Poisson(2)
0	0.1348	0.1353
1	0.2707	0.2707
2	0.2712	0.2707
3	0.1808	0.1804

The two agree to about three decimal places. The Poisson is the limit of the binomial as $n \to \infty$ and $p \to 0$ with np fixed; with $n = 500$ and $p = 0.004$ we are close to that limit. In practice this means we do not need to know n and p separately to model rare-event counts, only their product, the rate.

Solution 5.6 (Overdispersion)**.**

(a) No. A Poisson variable has variance equal to its mean, and here the variance is nearly three times the mean. With several years of monthly data, a ratio this large is not plausibly sampling noise. The counts are overdispersed.

(b) (i) Clustering: one incident raises the chance of others shortly after, for example cascading failures or a shared cause such as a storm. (ii) A varying rate: the underlying rate itself changes from month to month (seasonality, regime changes), and a mixture of Poissons with different means has variance larger than its mean.

(c) The Poisson model, with too little variance, puts too little probability in the tails. It would understate the chance of a bad month, potentially by a large factor. A negative binomial model (Chapter 16) allows the variance to exceed the mean.

Solution 5.7 (Memorylessness)**.**

(a) Mean $1/\lambda = 2$ years. The median solves $1 - e^{-0.5m} = 1/2$: $m = \ln 2/0.5 = 1.39$ years. Mean exceeds median because the distribution is right-skewed.

(b) $P(T > 3) = e^{-1.5} = 0.223$. $P(T > 5 \mid T > 2) = P(T > 5)/P(T > 2) = e^{-2.5}/e^{-1} = e^{-1.5} = 0.223$, the same.

(c) $P(T > s + t \mid T > t) = \dfrac{P(T > s + t)}{P(T > t)} = \dfrac{e^{-\lambda(s+t)}}{e^{-\lambda t}} = e^{-\lambda s} = P(T > s)$. (The event $T > s + t$ is contained in $T > t$, which justifies the first equality.)

(d) No. Memorylessness means a component that has run for ten years is as good as new, which contradicts wear. The exponential suits failures from external shocks arriving at a constant rate. Components that age need a distribution with an increasing failure rate, such as the Weibull or gamma with shape greater than one.

Solution 5.8 (Normal probabilities)**.** Standardize with $Z = (X - 50)/10$.

(a) $P(X > 65) = P(Z > 1.5) = 1 - 0.9332 = 0.0668$. By symmetry $P(X < 35) = P(Z < -1.5) = 0.0668$ as well.

(b) $P(-1 < Z < 1) = 2\Phi(1) - 1 = 0.6826$, the "68" of the 68–95–99.7 rule.

(c) $50 + 1.645(10) = 66.45$.

Solution 5.9 (Which is larger?)**.** $D = X - Y$ is a linear combination of independent normals, so it is normal with mean $10 - 6 = 4$ and variance $4 + 9 = 13$ (the variances add even though we subtract). Then

$$P(X < Y) = P(D < 0) = P\left(Z < \frac{0 - 4}{\sqrt{13}}\right) = \Phi(-1.11) = 1 - 0.8665 = 0.134.$$

Solution 5.10 (A lognormal price)**.**

(a) The median of $\ln S$ is 0.05, and because exp is increasing the median of S is $e^{0.05} = 1.0513$. The mean is $e^{\mu + \sigma^2/2} = e^{0.07} = 1.0725$. The mean exceeds the median because the lognormal is right-skewed: the convex exp stretches upside moves more than downside moves (Jensen again). A "typical" year returns about 5.1%, the expected return is 7.25%.

(b) $\mathrm{Var}(S) = (e^{0.04} - 1)e^{0.10 + 0.04} = 0.04081 \times 1.1503 = 0.0469$, so $\mathrm{SD}(S) = 0.217$.

(c) $P(S < 1) = P(\ln S < 0) = \Phi\big((0 - 0.05)/0.2\big) = \Phi(-0.25) = 1 - 0.5987 = 0.401$.

Solution 5.11 (Simulating by the inverse CDF)**.**

(a) For $x \geq 0$, $P(X \leq x) = P\big(-\ln(1-U) \leq \lambda x\big) = P\big(U \leq 1 - e^{-\lambda x}\big) = 1 - e^{-\lambda x}$, since $P(U \leq u) = u$ on $[0,1]$. This is the exponential CDF. Equivalently, $X = F^{-1}(U)$ where $F(x) = 1 - e^{-\lambda x}$.

(b) $-\ln(0.70)/2 = 0.178$ and $-\ln(0.05)/2 = 1.498$. Large uniforms map into the long right tail.

Solution 5.12 (The sampling distributions)**.**

(a) Mean 10, variance 20.

(b) $\nu/(\nu-2)$: $5/3 = 1.67$ for t_5 and $50/48 = 1.04$ for t_{50}, close to the standard normal's 1.

(c) $T = Z/\sqrt{V/\nu}$ with $Z \sim N(0,1)$ independent of $V \sim \chi^2_\nu$. Then $T^2 = \dfrac{Z^2/1}{V/\nu}$, and $Z^2 \sim \chi^2_1$ by the definition of the chi-squared. This is a ratio of independent chi-squareds, each divided by its degrees of freedom, so $T^2 \sim F_{1,\nu}$.

(d) $(n-1)s^2/\sigma^2 = 15s^2/4 \sim \chi^2_{15}$, so $s^2 = \frac{4}{15}\chi^2_{15}$. Hence $\mathbb{E}[s^2] = \frac{4}{15}(15) = 4 = \sigma^2$ (unbiased) and $\mathrm{Var}(s^2) = \frac{16}{225}(30) = 2.13$, so $\mathrm{SD}(s^2) = 1.46$. The upper 5% point is $\frac{4}{15}(25.00) = 6.67$: even with 16 observations, a sample variance two-thirds larger than the truth happens one time in twenty.

Solution 5.13 (How rare is a four-sigma day?)**.**

(a) $P(|Z| > 4) = 6.3 \times 10^{-5}$, about once every 15,800 days, or once every 63 years.

(b) t_3 has variance 3, so the rescaled variable exceeds 4 in absolute value when $|T| > 4\sqrt{3} = 6.93$. $P(|T| > 6.93) = 0.0062$: about once every 160 trading days, or 1.5 times a year.

(c) Both models have mean zero and variance one, so they agree on the "volatility," yet they disagree by a factor of about 100 on the frequency of a four-sigma day. A risk model calibrated only to the variance says nothing about the tails; the tail shape must be modeled separately. Empirical daily returns typically look much more like (b) than (a), which is the warning of Sections 2.3.2 and 5.2.3 in numbers.

Chapter 6

Sampling Distributions and the Central Limit Theorem

Solution 6.1 (Standard deviation or standard error?)**.**

(a) $\sigma/\sqrt{n} = 3.00/6 = \$0.50$.

(b) The standard deviation, \$3.00, describes individual days. The standard error, \$0.50, describes the uncertainty of the average.

(c) It attaches the day-to-day spread to an estimate of the mean, overstating the uncertainty of the average sixfold. The appropriate statement is $\$0.20 \pm \$0.50$ (one standard error), or $\pm$ about \$1.00 for a rough 95% interval, which shows that the average move is not distinguishable from zero. Conversely, quoting \$0.50 as the risk of holding a position for a day would understate that risk sixfold.

Solution 6.2 (Buying precision)**.**

(a) $12/\sqrt{n} \le 1$ requires $n \ge 144$.

(b) Halving requires four times the data, $n = 576$. A standard error of 0.1 requires $(12/0.1)^2 = 14{,}400$ observations, a hundred times the data for a tenfold gain. Precision grows only like $\sqrt{n}$.

Solution 6.3 (The CLT for a mean)**.**

(a) $\bar{X}$ is approximately $N(100,\, 15^2/36)$, so its standard error is 2.5 and $P(\bar{X} > 104) \approx P(Z > 1.6) = 1 - 0.9452 = 0.055$.

(b) The CLT says the sampling distribution of $\bar{X}$ is approximately normal whatever the population's shape, provided the variance is finite and n is reasonably large. Only μ and σ enter.

(c) Not without more information. With $n = 4$ the shape of the population still shows through in $\bar{X}$, so the normal approximation is reliable only if the population is itself close to normal. The mean and standard error (100 and 7.5) are still exact; only the normal shape is in doubt.

Solution 6.4 (The CLT for a total)**.** The total S has mean $50 \times 200 = \$10{,}000$ and variance 50×80^2, so $\mathrm{SD}(S) = 80\sqrt{50} = \$565.7$. Note that the standard deviation of a total grows like $\sqrt{n}$, while that of an average shrinks like $1/\sqrt{n}$. By the CLT,

$$P(S > 11{,}000) \approx P\left(Z > \frac{1{,}000}{565.7}\right) = P(Z > 1.77) = 0.038.$$

Solution 6.5 (The CLT for a proportion)**.**

(a) $\sqrt{0.3 \times 0.7/200} = 0.0324$.

(b) $P(\hat{p} > 0.35) \approx P\big(Z > (0.35 - 0.30)/0.0324\big) = P(Z > 1.54) = 1 - 0.9382 = 0.062$.

(c) $np = 60$ and $n(1-p) = 140$, both well above 10, so the approximation is reasonable. It is still an approximation for a discrete count: the exact binomial probability that more than 70 of the 200 switch is 0.054. Applying a continuity correction (using $70.5/200$ as the cutoff) brings the normal answer to 0.053.

Solution 6.6 (When the normal approximation fails)**.**

(a) $0.99^{100} = 0.366$.

(b) The standard error is $\sqrt{0.0099/100} = 0.00995$, so $P(\hat{p} \leq 0) \approx \Phi(-0.01/0.00995) = \Phi(-1.005) = 0.157$.

(c) The approximation is off by more than a factor of two. Here $np = 1$, far below the threshold of about 10. The count of defects is concentrated on $0, 1, 2$ and has a long right tail; a symmetric normal curve centered at 1 defect cannot represent a distribution that piles up against the boundary at zero. Use the exact binomial, or its Poisson approximation $e^{-1} = 0.368$.

Solution 6.7 (Which theorem?)**.**

(a) LLN: a frequency is the sample mean of $0/1$ indicators.

(b) CLT (assuming finite variance and independence).

(c) Neither, and the statement is false. A large sample makes the histogram a better picture of the population, which is skewed; the CLT concerns the distribution of the *mean*, not of the data.

(d) LLN.

(e) Neither. The Cauchy distribution has no finite variance (indeed no mean), and the average of independent Cauchy variables has the same Cauchy distribution as a single one. A simulation of 1,000-observation averages gives an interquartile range of about $(-1, 1)$, the same as for one draw. Averaging achieves nothing.

Solution 6.8 (Sample variances vary)**.** Since $10s^2/25 \sim \chi^2_{10}$, we have $s^2 = 2.5\,\chi^2_{10}$. The lower 5% point is $2.5(3.940) = 9.85$ and the upper 5% point is $2.5(18.307) = 45.77$. So 90% of samples of 11 give a sample variance between about 10 and 46 when the truth is 25: from about 60% too low to 83% too high. The interval is lopsided because the chi-squared is right-skewed. Variances, and hence volatilities, are much harder to pin down from small samples than intuition suggests.

Solution 6.9 (Why the t distribution matters)**.**

(a) $2(1 - \Phi(2.776)) = 0.0055$, about ten times smaller than the true 0.05.

(b) Those intervals miss the true mean 12.2% of the time, not 5%: their real confidence level is about 88%. The extra uncertainty from estimating σ by s with only four degrees of freedom is large, and ignoring it produces overconfident conclusions.

(c) With 29 degrees of freedom the error is small (6% rather than 5%), which is why the z and t procedures nearly coincide for moderate and large samples. Using the t critical value is never wrong when σ is estimated, so there is no reason to switch.

Solution 6.10 (How large must n be?)**.**

(a) $2/\sqrt{5} = 0.89$, $2/\sqrt{30} = 0.37$, $2/\sqrt{100} = 0.20$.

(b) $2/\sqrt{n} = 0.2$ gives $n = 100$.

(c) With 200,000 simulated samples, the upper tail probability is about 0.034 and the lower about 0.014. Their sum is close to the nominal 0.05, but the split is badly unbalanced: the right tail is too heavy and the left too light. A one-sided test at the 2.5% level would have an actual error rate of 3.4% in one direction and 1.4% in the other. "$n \geq 30$" is not a guarantee for a skewed population; for an exponential parent it takes n in the hundreds before the two tails come into balance.

Chapter 7

Estimation and Confidence Intervals

Solution 7.1 (Shrinking an estimator)**.**

(a) T_1: bias 0, variance $\sigma^2/n = 0.4$, MSE 0.4. T_2: $\mathbb{E}[T_2] = 0.8\mu$, so bias -0.2μ; variance $0.64 \times 0.4 = 0.256$; MSE $0.256 + 0.04\mu^2$.

(b) At $\mu = 2$: $\text{MSE}(T_2) = 0.256 + 0.16 = 0.416 > 0.4$, so T_1 wins. At $\mu = 1$: $0.256 + 0.04 = 0.296 < 0.4$, so T_2 wins by a quarter.

(c) $0.256 + 0.04\mu^2 < 0.4$ when $\mu^2 < 3.6$, i.e. $|\mu| < 1.90$. Shrinkage toward zero pays when the truth is near zero and costs when it is far away. Since μ is unknown, the choice embodies a judgment about where μ is likely to be, the same judgment a Bayesian prior makes explicit (Chapter 17) and regularization tunes by cross-validation (Chapter 18).

Solution 7.2 (The MSE decomposition)**.** Write $\hat\theta - \theta = (\hat\theta - \mathbb{E}\hat\theta) + (\mathbb{E}\hat\theta - \theta)$. The second term is a constant b, the bias. Squaring and taking expectations,

$$\mathbb{E}[(\hat\theta - \theta)^2] = \mathbb{E}[(\hat\theta - \mathbb{E}\hat\theta)^2] + 2b\,\mathbb{E}[\hat\theta - \mathbb{E}\hat\theta] + b^2 = \text{Var}(\hat\theta) + 0 + b^2,$$

because $\mathbb{E}[\hat\theta - \mathbb{E}\hat\theta] = 0$. It is the same add-and-subtract argument as Exercise 2.6(b).

Solution 7.3 (Should we divide by n?)**.**

(a) $\mathbb{E}[\hat\sigma^2] = \frac{n-1}{n}\sigma^2$, so the bias is $-\sigma^2/n$. $\text{Var}(\hat\sigma^2) = \frac{(n-1)^2}{n^2} \cdot \frac{2\sigma^4}{n-1} = \frac{2(n-1)\sigma^4}{n^2}$. Adding the squared bias σ^4/n^2 gives $\text{MSE} = (2n-1)\sigma^4/n^2$.

(b) We need $(2n-1)/n^2 < 2/(n-1)$, i.e. $(2n-1)(n-1) < 2n^2$, i.e. $2n^2 - 3n + 1 < 2n^2$, which holds for $n \geq 1$. For $n = 5$: $\text{MSE}(s^2) = 0.50\,\sigma^4$ and $\text{MSE}(\hat\sigma^2) = 0.36\,\sigma^4$.

(c) Bessel's correction buys unbiasedness, not accuracy. The biased n-divisor has smaller mean squared error for normal data (dividing by $n+1$ does better still). Which to use depends on the goal; $n-1$ remains the convention because unbiasedness is convenient in the t, F and regression formulas built on s^2, and for large n the difference is negligible.

Solution 7.4 (Method of moments)**.**

(a) Set $\theta/2 = \bar x$: $\hat\theta = 2\bar x$.

(b) $\bar x = 12.5/5 = 2.5$, so $\hat\theta = 5.0$, comfortably above the largest observation 4.1.

(c) $\bar x = 5.8/4 = 1.45$, so $\hat\theta = 2.9$. But the data include 4.0, which is impossible if $\theta = 2.9$. The method of moments uses only the mean and ignores the constraint that every observation must be below θ. (The maximum likelihood estimator here is the sample maximum, which never makes this mistake, although it is biased slightly low.)

Solution 7.5 (MLE for an exponential rate)**.**

(a) $\ell(\lambda) = \sum_i \big(\ln\lambda - \lambda x_i\big) = n\ln\lambda - \lambda\sum_i x_i$. Setting $\ell'(\lambda) = n/\lambda - \sum x_i = 0$ gives $\hat\lambda = n/\sum x_i = 1/\bar x$. Since $\ell''(\lambda) = -n/\lambda^2 < 0$, this is a maximum.

(b) $\sum x_i = 15$ and $n = 5$, so $\hat\lambda = 1/3$ per minute: one arrival every three minutes on average, matching $\bar x = 3$.

Solution 7.6 (MLE for a Poisson rate)**.**

(a) $\ell(\lambda) = \sum_i \left(x_i \ln \lambda - \lambda - \ln x_i!\right) = (\sum x_i)\ln\lambda - n\lambda + \text{const}$. Then $\ell'(\lambda) = \sum x_i/\lambda - n = 0$ gives $\hat\lambda = \bar x$, and $\ell'' < 0$ confirms a maximum.

(b) $\bar x = 12/6 = 2$.

Solution 7.7 (MLE for the normal)**.** The log-likelihood is $\ell(\mu, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\partial\ell/\partial\mu = \frac{1}{\sigma^2}\sum(x_i - \mu) = 0$ gives $\hat\mu = \bar x$. Setting $\partial\ell/\partial\sigma^2 = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum(x_i - \hat\mu)^2 = 0$ gives $\hat\sigma^2 = \frac{1}{n}\sum(x_i - \bar x)^2$.

The MLE of σ^2 divides by n and is biased downward by the factor $(n-1)/n$, as in Exercise 7.3. Maximum likelihood guarantees consistency and asymptotic efficiency, not unbiasedness in finite samples.

Solution 7.8 (A z-interval)**.** The standard error is $12/6 = 2$. 90%: $101.5 \pm 1.645(2) = (98.21,\ 104.79)$. 95%: $101.5 \pm 1.96(2) = (97.58,\ 105.42)$. 99%: $101.5 \pm 2.576(2) = (96.35,\ 106.65)$.

Demanding more confidence widens the interval: the 99% interval is about 57% wider than the 90% one. With the data fixed, more confidence can be had only by saying less.

Solution 7.9 (A t-interval)**.**

(a) $\bar x = 505/10 = 50.5$. The squared deviations sum to 82.5, so $s^2 = 82.5/9 = 9.17$ and $s = 3.03$. The standard error is $3.03/\sqrt{10} = 0.957$.

(b) $50.5 \pm 2.262(0.957) = 50.5 \pm 2.17 = (48.33,\ 52.67)$.

(c) $50.5 \pm 1.96(0.957) = (48.62,\ 52.38)$, about 13% narrower. With nine degrees of freedom such "95%" intervals actually cover only about 92% of the time.

Solution 7.10 (An interval for a proportion)**.** $\hat p = 420/1200 = 0.35$, with standard error $\sqrt{0.35(0.65)/1200} = 0.0138$. The interval is $0.35 \pm 1.96(0.0138) = 0.35 \pm 0.027 = (0.323,\ 0.377)$.

The 95% describes the procedure: if we repeated the random survey many times and built an interval each time, about 95% of those intervals would contain the true proportion. It is not the probability that this particular interval contains it. The interval also accounts only for sampling error; nonresponse or a biased sampling frame (Exercise 1.6) would not be reflected in its width.

Solution 7.11 (An interval for a variance)**.** From $P\left(2.700 \le 9s^2/\sigma^2 \le 19.023\right) = 0.95$, invert:

$$\frac{9s^2}{19.023} \le \sigma^2 \le \frac{9s^2}{2.700} \quad\Longrightarrow\quad (4.26,\ 30.0).$$

Taking square roots, $\sigma \in (2.06,\ 5.48)$, against the point estimate $s = 3$.

The chi-squared distribution is right-skewed, so its percentage points are not symmetric about its mean, and inverting puts the long side of the interval above s^2. The interval is also very wide: ten observations pin down a variance only to within a factor of seven. This interval relies heavily on normality and is unreliable for heavy-tailed data.

Solution 7.12 (Planning a sample size)**.**

(a) $n = (1.96 \times 15/2)^2 = 216.1$, so round up to 217. Halving the margin to 1 gives 864.4, so 865: four times as many.

(b) The margin is $1.96\sqrt{p(1-p)/n}$, and $p(1-p) \le 1/4$ with equality at $p = 1/2$. The safe choice is $n = 1.96^2(0.25)/0.03^2 = 1067.1$, so 1,068, which is why national polls use about a thousand respondents. If $p \approx 0.1$, $p(1-p) = 0.09$ and $n = 385$ suffices.

Solution 7.13 (What does the interval mean?)**.**

(a) Incorrect in the frequentist framework: μ is fixed, and this interval either contains it or not. (A Bayesian credible interval, Chapter 17, does support this reading.)

(b) Correct: this is the definition.

(c) Incorrect: the interval concerns the mean, not individual observations, which are far more spread out.

(d) Incorrect. The interval is centered on this study's $\bar{x}$, not on μ. Averaged over possible first studies, the chance that a second sample mean lands inside the first interval is only about 83%, because the difference of two independent means has variance $2\sigma^2/n$ and $P(|Z| < 1.96/\sqrt{2}) = 0.83$. It reaches 95% only if the first $\bar{x}$ happened to equal μ exactly.

(e) Correct, by the duality of tests and intervals (Section 8.3.1).

Chapter 8

Hypothesis Testing

Solution 8.1 (Setting up the hypotheses)**.**

(a) $H_0 : \mu = 2$ (or $\mu \leq 2$) against $H_1 : \mu > 2$; one-sided, because only a longer delay is of concern and the direction was fixed before seeing data.

(b) $H_0 : \mu = 0$ against $H_1 : \mu \neq 0$; two-sided, because bias in either direction matters.

(c) H_0 : no change in mean demand against H_1 : some change; two-sided, unless a direction was specified in advance on substantive grounds.

(d) Choosing the direction after looking at the data doubles the real Type I error rate: in effect the analyst is running a two-sided test at level 2α while reporting level α. The alternative must be fixed before the data are seen.

Solution 8.2 (Two kinds of error)**.**

(a) Type I: the alarm fires when there is no leak (a false alarm). Type II: there is a leak and the alarm stays silent (a missed detection).

(b) Usually a missed leak is far costlier than a false alarm, which argues for a relatively large α to gain power, accepting more false alarms. The conventional 0.05 is a convention, not a law; α should reflect the costs of the two errors. (Too many false alarms carry their own cost: operators learn to ignore the alarm.)

(c) More information per decision: more or better sensors, longer averaging windows, or lower measurement noise. Power rises with sample size and with signal relative to noise.

Solution 8.3 (A z-test)**.**

(a) $z = (52.1 - 50)/(8/\sqrt{64}) = 2.1/1 = 2.1$. Two-sided $p = 2(1 - 0.9821) = 0.036$.

(b) Reject H_0 at 5% ($0.036 < 0.05$); do not reject at 1%. The evidence is moderate, not overwhelming.

(c) $1 - 0.9821 = 0.018$, half the two-sided value, legitimate only if the one-sided alternative was chosen in advance.

Solution 8.4 (A t-test and its interval)**.**

(a) $t = (50.5 - 48)/0.957 = 2.61 > 2.262$: reject. (Exact $p = 0.028$.)

(b) $t = (50.5 - 52)/0.957 = -1.57$, and $|t| < 2.262$: do not reject. ($p = 0.15$.)

(c) 48 lies outside $(48.33, 52.67)$ and is rejected; 52 lies inside and is not. The interval is exactly the set of null values a two-sided 5% test would not reject.

Solution 8.5 (Paired or unpaired?)**.**

(a) The differences are $1.5, 1.0, 1.8, 1.1, 1.4, 1.2$, with mean $\bar{d} = 1.333$ and standard deviation $s_d = 0.294$. The standard error is $0.294/\sqrt{6} = 0.120$, so $t = 1.333/0.120 = 11.1$, far beyond 2.571 ($p = 0.0001$). The modification clearly raised the measurements.

(b) The two columns have standard deviations 9.35 and 9.31, so the standard error of the difference in means is $\sqrt{9.35^2/6 + 9.31^2/6} = 5.39$ and $t = 1.333/5.39 = 0.25$ ($p = 0.81$). Nothing is detected.

(c) The units differ enormously from one another (20 to 45), and that between-unit variation swamps the small, consistent effect when the samples are treated as independent. Pairing compares each unit with itself, which removes the between-unit variation entirely and leaves only the variability of the change. The same data give $t = 11$ or $t = 0.25$ depending on whether the design is respected.

Solution 8.6 (Power)**.** Under $\mu = 103$ the test statistic $z = (\bar{X} - 100)/(10/\sqrt{n})$ is normal with mean $3\sqrt{n}/10$ and variance 1, so the power is $P(Z > 1.645 - 3\sqrt{n}/10)$.

(a) $n = 25$: the mean shift is 1.5, so the power is $P(Z > 0.145) = 1 - 0.5576 = 0.442$.

(b) $n = 100$: the shift is 3, so the power is $P(Z > -1.355) = 0.912$.

(c) Require $1.645 - 3\sqrt{n}/10 = -0.842$: $\sqrt{n} = 8.29$, so $n = 68.7$; take 69.

(d) Very little. With power 0.44 the test misses a real effect of this size more often than it finds it. A non-significant result from an underpowered study is weak evidence of no effect; report the confidence interval, which will be wide and include both 100 and 103.

Solution 8.7 (Reading a p-value)**.** Only (b) and (d) are correct. (a) and (e) commit the inversion fallacy: the p-value is $P(\text{data this extreme} \mid H_0)$, not $P(H_0 \mid \text{data})$; the latter requires a prior (Chapters 3 and 17). (c) A p-value mixes effect size with sample size; a tiny effect gives a small p in a large sample (Exercise 8.12). (f) Selective reporting from many tests invalidates the nominal error rate (Exercise 8.8).

Solution 8.8 (Many tests)**.**

(a) Expected number $20 \times 0.05 = 1$. $P(\text{at least one}) = 1 - 0.95^{20} = 0.642$.

(b) Each test uses 0.0025, so $P(\text{at least one}) = 1 - 0.9975^{20} = 0.049$, just under 0.05. (The union bound of Exercise 3.13 guarantees at most 0.05 even when the tests are dependent.)

(c) Power. Each test now needs much stronger evidence, so real effects are missed more often. With many tests, the honest alternatives are to pre-specify a few primary hypotheses, to control the false discovery rate instead, or to confirm findings on fresh data.

Solution 8.9 (Is the die fair?)**.** Under H_0 each expected count is 20. Then

$$\chi^2 = \frac{(-5)^2 + 5^2 + (-2)^2 + 2^2 + (-6)^2 + 6^2}{20} = \frac{130}{20} = 6.5,$$

on $6 - 1 = 5$ degrees of freedom. Since $6.5 < 11.07$ we do not reject ($p \approx 0.26$). The counts look uneven, but unevenness of this size is common with 120 rolls of a fair die. This does not prove the die fair; a small bias would need many more rolls to detect.

Solution 8.10 (A test of independence)**.** Expected counts under independence are (row total $\times$ column total)/250:

	spike	no spike
high	$60 \cdot 27/250 = 6.48$	53.52
calm	20.52	169.48

All expected counts exceed 5, so the chi-squared approximation is reasonable. Then

$$\chi^2 = \frac{11.52^2}{6.48} + \frac{11.52^2}{53.52} + \frac{11.52^2}{20.52} + \frac{11.52^2}{169.48} = 20.48 + 2.48 + 6.47 + 0.78 = 30.2,$$

on $(2-1)(2-1) = 1$ degree of freedom. This far exceeds 3.841 ($p \approx 4 \times 10^{-8}$): spikes are strongly associated with the high-volatility regime. Note that every cell has the same absolute discrepancy 11.52; that is always true in a 2×2 table, because the margins are fixed.

Solution 8.11 (Comparing variances)**.** $F = s_1^2/s_2^2 = 2.5$ on $(12, 15)$ degrees of freedom. Since $0.315 < 2.5 < 2.96$, do not reject ($p \approx 0.10$). A variance ratio of 2.5 is not conclusive with samples this small, echoing Exercise 6.8. The F-test assumes both populations are normal and is quite sensitive to violations, especially heavy tails, which inflate its Type I error. Robust alternatives such as Levene's test are preferred in practice.

Solution 8.12 (Significant but trivial)**.**

(a) The standard error is $1/\sqrt{10^6} = 0.001$, so $z = 0.01/0.001 = 10$; the p-value is about 10^{-23}.

(b) $0.01 \pm 1.96(0.001) = (0.008,\ 0.012)$.

(c) Statistically the change is established beyond doubt; practically it is one hundredth of a standard deviation. Whether that matters is a question about the business, not about statistics, and the confidence interval answers it far better than the p-value: the effect is precisely estimated to be tiny.

Solution 8.13 (Comparing two strategies)**.**

(a) The standard error of the difference in means is

$$\sqrt{\frac{1.8^2}{50} + \frac{2.1^2}{50}} = 0.391,$$

so $z = 0.7/0.391 = 1.79$. Two-sided $p = 2(1 - 0.9633) = 0.073$. Do not reject at 5%.

(b) $0.7 \pm 1.96(0.391) = (-0.07,\ 1.47)$.

(c) Failing to reject is not evidence of equality. The interval runs from essentially no difference to a difference of 1.47, a third of A's mean. The data are compatible with a substantial advantage for A; they simply do not establish it. "No significant difference" and "the same" are different claims.

Chapter 9

Covariance and Correlation

Solution 9.1 (Covariance and correlation by hand). $\bar{x} = 3$ and $\bar{y} = 4$. The deviations are x: $-2, -1, 0, 1, 2$ and y: $-2, 0, 1, 0, 1$. The products are $4, 0, 0, 0, 2$, summing to 6, so $s_{XY} = 6/4 = 1.5$. The squared deviations sum to 10 for x and 6 for y, so $s_X^2 = 2.5$ and $s_Y^2 = 1.5$. Then

$$r = \frac{1.5}{\sqrt{2.5 \times 1.5}} = \frac{1.5}{1.936} = 0.775.$$

These data reappear in Exercises 10.1 and 11.1.

Solution 9.2 (Units and linear changes).

(a) The covariance is multiplied by 100, to 150; the correlation is unchanged at 0.775. The size of a covariance reflects units; the correlation does not.

(b) By bilinearity, $\text{Cov}(2X + 1, -3Y + 4) = 2(-3)\,\text{Cov}(X, Y) = -6(1.5) = -9$. Each standard deviation is multiplied by the absolute value of its factor, so Corr $= \frac{-6}{|2||-3|}\, r = -r = -0.775$. A linear change can flip the sign of a correlation but not its magnitude.

Solution 9.3 (Perfect dependence, zero correlation). $\bar{x} = 0$ and $\bar{y} = 2$. The products of deviations $x_i(y_i - 2)$ are $-4, 1, 0, -1, 4$, summing to 0, so $r = 0$. The scatterplot shows a perfect U: y is completely determined by x, but the rising right half cancels the falling left half. The same cancellation gives $\text{Corr}(x, |x|) = 0$. Zero correlation means no *linear* relationship, nothing more.

Solution 9.4 (Variance of a sum). Cov $= -0.5(10)(20) = -100$, so $\text{Var}(X + Y) = 100 + 400 - 200 = 300$ and SD $= 17.3$. With $\rho = 0$: Var $= 500$, SD $= 22.4$. With $\rho = 1$: Var $= 900$, SD $= 30$, the only case in which standard deviations add. Negative correlation removes a large part of the risk.

Solution 9.5 (The minimum-variance hedge).

(a) $V(h) = \sigma_S^2 - 2h\,\text{Cov}(\Delta S, \Delta F) + h^2\sigma_F^2$. Setting $V'(h) = -2\,\text{Cov} + 2h\sigma_F^2 = 0$ gives $h^* = \text{Cov}\,/\sigma_F^2 = \rho\sigma_S/\sigma_F$; $V'' > 0$ confirms a minimum. Substituting, $V(h^*) = \sigma_S^2 - \text{Cov}^2\,/\sigma_F^2 = \sigma_S^2(1 - \rho^2)$.

(b) $h^* = 0.9(3/2.5) = 1.08$ units of futures per unit of spot. The hedged standard deviation is $3\%\sqrt{1 - 0.81} = 1.31\%$, a 56% reduction.

(c) $\sqrt{1 - \rho^2}$: 0.87, 0.71, and 0.31. Hedging is weak unless the correlation is high: even at $\rho = 0.7$, more than 70% of the risk remains. This is basis risk. Note that h^* is exactly the slope of the regression of ΔS on ΔF (Chapter 10), and ρ^2 is the fraction of variance the hedge removes.

Solution 9.6 (One outlier, two correlations).

(a) Pearson $r = 0.72$. The y values are in the same order as the x values, so the ranks agree perfectly and Spearman $= 1$.

(b) Pearson $r = 0.997$. The single outlier at 40 lowered Pearson's r from nearly 1 to 0.72, although it continues the upward trend.

(c) Pearson $r = -0.44$: one point reverses the sign. The y ranks become $2, 3, 4, 5, 6, 1$, and Spearman's correlation is 0.14: lowered, but still positive. Pearson's r is built on means and squared deviations and inherits their sensitivity; the rank correlation is robust in the same way as the median.

Solution 9.7 (Monotone but curved)**.** Pearson $r = 0.89$; Spearman $= 1$. The relationship is perfectly monotone, so the rank correlation is exactly one. Pearson's r measures closeness to a straight line, and an exponential curve is not one. (After taking logs of y the Pearson correlation would be exactly 1.)

Solution 9.8 (What explains the correlation?)**.**

(a) Confounding by city size, and to some extent reverse causation within a city: bigger fires bring more firefighters.

(b) Confounding by temperature, which drives both.

(c) Probably both directions. Marketing may attract money, but funds with strong recent performance and growing assets both attract inflows and have bigger marketing budgets. An experiment or a careful design is needed to separate them.

(d) Chance (data mining). Among 500 unrelated indicators some will correlate by luck (Exercise 9.10); the finding needs confirmation on new data.

(e) Confounding by patient severity: hospitals that treat sicker patients need more ICU capacity and have higher mortality.

(f) Largely a shared driver, not a direct effect of gas generation on price: high demand both brings expensive gas units online and sets higher prices because those units are on the margin. Here the mechanism is well understood, but the correlation alone would not reveal it.

Solution 9.9 (Why $|\rho| \le 1$)**.** $\mathrm{Var}\left(X/\sigma_X \pm Y/\sigma_Y\right) = 1 + 1 \pm 2\,\mathrm{Cov}(X, Y)/(\sigma_X\sigma_Y) = 2(1 \pm \rho)$. Variances are non-negative, so $1 + \rho \ge 0$ and $1 - \rho \ge 0$, i.e. $-1 \le \rho \le 1$. Equality $\rho = 1$ holds exactly when $\mathrm{Var}(X/\sigma_X - Y/\sigma_Y) = 0$, i.e. $X/\sigma_X - Y/\sigma_Y$ is a constant: Y is an exact increasing linear function of X. Similarly $\rho = -1$ means an exact decreasing linear relationship.

Solution 9.10 (Screening for correlations)**.**

(a) The standard deviation is 0.1, so $P(|r| > 0.2) \approx P(|Z| > 2) = 0.046$.

(b) About $100 \times 0.046 \approx 4.6$, so four or five spurious "findings" of $|r| > 0.2$ are expected even though none of the predictors is related to the target.

(c) Treat it as a hypothesis, not a result: adjust for the number of candidates screened (Chapter 8, multiple comparisons), and test the relationship on data not used in the screen (out-of-sample validation, Chapters 11 and 18). A plausible mechanism also helps.

Chapter 10

Simple Linear Regression

Solution 10.1 (Fitting a line by hand)**.**

(a) From Exercise 9.1, $\sum(x_i - \bar{x})(y_i - \bar{y}) = 6$ and $\sum(x_i - \bar{x})^2 = 10$, so $\hat{\beta}_1 = 0.6$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 4 - 0.6(3) = 2.2$. The fitted line is $\hat{y} = 2.2 + 0.6x$.

(b) Fitted values: 2.8, 3.4, 4.0, 4.6, 5.2. Residuals $y - \hat{y}$: -0.8, 0.6, 1.0, -0.6, -0.2, which sum to 0.

(c) $\text{SST} = \sum(y_i - 4)^2 = 4 + 0 + 1 + 0 + 1 = 6$. $\text{SSE} = 0.64 + 0.36 + 1 + 0.36 + 0.04 = 2.4$. $\text{SSR} = \sum(\hat{y}_i - 4)^2 = 1.44 + 0.36 + 0 + 0.36 + 1.44 = 3.6$, and indeed $3.6 + 2.4 = 6$. $R^2 = 3.6/6 = 0.60$, which equals $r^2 = 0.775^2$.

Solution 10.2 (Inference on the slope)**.**

(a) $s = \sqrt{\text{SSE}/(n-2)} = \sqrt{2.4/3} = 0.894$. $\text{se}(\hat{\beta}_1) = s/\sqrt{\sum(x_i - \bar{x})^2} = 0.894/\sqrt{10} = 0.283$.

(b) $t = 0.6/0.283 = 2.12 < 3.182$: do not reject ($p = 0.12$). The interval is $0.6 \pm 3.182(0.283) = (-0.30,\ 1.50)$, which includes zero.

(c) $F = \text{SSR}/(\text{SSE}/3) = 3.6/0.8 = 4.5 = 2.12^2$.

(d) With five points and three residual degrees of freedom, the critical value is large (3.18 rather than 1.96) and the standard error is large. R^2 describes the fit in this sample; it says nothing about how reliably that fit would recur. Five points can produce a fairly high R^2 by chance.

Solution 10.3 (The normal equations)**.**

(a) $\partial\,\text{SSE}/\partial b_0 = -2\sum(y_i - b_0 - b_1x_i) = -2\sum e_i = 0$ and $\partial\,\text{SSE}/\partial b_1 = -2\sum x_i(y_i - b_0 - b_1x_i) = -2\sum x_ie_i = 0$.

(b) Dividing $\sum e_i = 0$ by n: $\bar{y} - \hat{\beta}_0 - \hat{\beta}_1\bar{x} = 0$, so $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$ and the line passes through $(\bar{x}, \bar{y})$. Substituting into the second equation, $\sum x_i\big[(y_i - \bar{y}) - \hat{\beta}_1(x_i - \bar{x})\big] = 0$. Since $\sum \bar{x}\big[(y_i - \bar{y}) - \hat{\beta}_1(x_i - \bar{x})\big] = 0$, we may replace x_i by $x_i - \bar{x}$, giving $\sum(x_i - \bar{x})(y_i - \bar{y}) = \hat{\beta}_1\sum(x_i - \bar{x})^2$.

The two conditions say the residuals are uncorrelated with the predictor (and hence with the fitted values), which is why SST splits exactly into SSR plus SSE.

Solution 10.4 (The line from summary statistics)**.** $\hat{\beta}_1 = r\,s_Y/s_X = 0.6(10/4) = 1.5$ and $\hat{\beta}_0 = 50 - 1.5(20) = 20$, so $\hat{y} = 20 + 1.5x$. At $x = 25$: 57.5. $R^2 = r^2 = 0.36$.

Solution 10.5 (Regressing the other way)**.**

(a) Slope $r\,s_X/s_Y = 0.6(4/10) = 0.24$; intercept $20 - 0.24(50) = 8$. So $\hat{x} = 8 + 0.24y$.

(b) $\hat{x} = 8 + 0.24(60) = 22.4$. Inverting the first line gives $(60 - 20)/1.5 = 26.7$.

(c) $1.5 \times 0.24 = 0.36 = r^2$. Each regression predicts its outcome to be fewer standard deviations from its mean than the predictor is: one standard deviation in Y predicts only $r = 0.6$ standard deviations in X, and vice versa. This is regression to the mean. The two lines minimize different things (vertical versus horizontal squared distances) and coincide only when the points lie exactly on a line. Regression is asymmetric: decide which variable is being predicted before fitting.

Solution 10.6 (Interpreting a fitted line)**.**

(a) Each additional degree of maximum temperature is associated with 0.55 GW more expected peak load, within the summer range studied.

(b) Formally it is the predicted load at 0°F, -12 GW, which is impossible. Zero is far outside the data, so the intercept only anchors the line; it has no direct interpretation.

(c) $-12.0 + 0.55(90) = 37.5$ GW.

(d) At 110°F, 48.5 GW, cautiously: it is a modest extrapolation, but the load–temperature relationship often steepens or saturates at extremes, and the prediction interval widens with distance from $\bar{x}$. At 20°F the equation predicts -1 GW, which is absurd: in winter, load *rises* as temperature falls because of heating. The linear relationship holds only over the range where it was estimated.

Solution 10.7 (A slope test from regression output)**.**

(a) $t = 2.5/0.8 = 3.125 > 2.101$: reject ($p \approx 0.006$).

(b) $2.5 \pm 2.101(0.8) = (0.82,\ 4.18)$.

(c) $t = (2.5 - 1)/0.8 = 1.875 < 2.101$: do not reject. Consistently, 1 lies inside the interval. The data show the slope is positive but cannot distinguish 1 from 4.

Solution 10.8 (Confidence versus prediction intervals)**.**

(a) $\hat{y} = 4.0$. Mean response: se $= 0.894\sqrt{1/5 + 0} = 0.400$, interval $4.0 \pm 1.27 = (2.73,\ 5.27)$. New observation: se $= 0.894\sqrt{1 + 1/5} = 0.980$, interval $4.0 \pm 3.12 = (0.88,\ 7.12)$.

(b) $\hat{y} = 5.8$. Mean response: se $= 0.894\sqrt{1/5 + 9/10} = 0.938$, interval $(2.81,\ 8.79)$. New observation: se $= 0.894\sqrt{1 + 1/5 + 9/10} = 1.296$, interval $(1.68,\ 9.92)$.

The prediction interval is much wider because it includes the scatter of an individual point about the line, which no amount of data removes. Both intervals widen away from $\bar{x}$; at $x_0 = 6$, outside the observed range, the confidence interval for the mean is more than twice as wide as at the center. And all of this assumes the line remains valid at $x_0 = 6$, which the data cannot confirm.

Solution 10.9 (Where to put the x-values)**.**

(a) Equal spacing over an interval of width w with $n = 10$ gives $\sum(x_i - \bar{x})^2 = w^2 \cdot \frac{n(n+1)}{12(n-1)}$. Design A ($w = 2$): 4.07. Design B ($w = 10$): 101.9.

(b) $\text{se}(\hat{\beta}_1) \propto 1/\sqrt{\sum(x_i - \bar{x})^2}$, so design B's standard error is $\sqrt{101.9/4.07} = 5$ times smaller. Spreading the design is equivalent to having 25 times as much data.

(c) Design B assumes the relationship is linear over the whole range from 0 to 10. If it curves, the wider design gives a precise estimate of a misspecified line. A common compromise is to spread the design but include a few points in the middle to check for curvature.

Solution 10.10 (High R^2, wrong model)**.**

(a) $\hat{y} = -15 + 10x$, with $R^2 = 0.93$.

(b) The residuals are $15, 6, -1, -6, -9, -10, -9, -6, -1, 6, 15$: positive at both ends, negative in the middle, a U-shape. Residuals that follow a smooth pattern against x are the signature of a wrong functional form (Chapter 12).

(c) R^2 measures how much variation the line accounts for, not whether the line is the right model. Here the relationship is exact and the linear model is wrong everywhere, yet R^2 is 0.93. Always look at the residuals.

Solution 10.11 (Which assumption buys what?)**.**

(a) Linearity with zero-mean errors given x (no omitted variables correlated with x). Nothing else.

(b) Linearity, homoskedasticity, and uncorrelated errors: the Gauss–Markov conditions. Normality is not needed.

(c) Homoskedasticity and uncorrelated errors, on top of linearity. Without them, OLS is still unbiased but the formula is wrong (Chapter 12).

(d) All four, including normality.

(e) Linearity, homoskedasticity, and independence; normality is replaced by the Central Limit Theorem, provided the error variance is finite.

Chapter 11

Multiple Linear Regression

Solution 11.1 (Least squares in matrix form)**.**

(a) $\mathbf{X}$ has a column of five ones and the column $(1,2,3,4,5)^\top$.

$$\mathbf{X}^\top\mathbf{X} = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix} = \begin{pmatrix} 5 & 15 \\ 15 & 55 \end{pmatrix}, \qquad \mathbf{X}^\top\mathbf{y} = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix} = \begin{pmatrix} 20 \\ 66 \end{pmatrix}.$$

(b) The determinant is $5(55) - 15^2 = 50$, so

$$(\mathbf{X}^\top\mathbf{X})^{-1} = \frac{1}{50}\begin{pmatrix} 55 & -15 \\ -15 & 5 \end{pmatrix} = \begin{pmatrix} 1.1 & -0.3 \\ -0.3 & 0.1 \end{pmatrix},$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y} = \begin{pmatrix} 1.1(20) - 0.3(66) \\ -0.3(20) + 0.1(66) \end{pmatrix} = \begin{pmatrix} 2.2 \\ 0.6 \end{pmatrix},$$

the same as Exercise 10.1.

(c) $s^2(\mathbf{X}^\top\mathbf{X})^{-1} = \begin{pmatrix} 0.88 & -0.24 \\ -0.24 & 0.08 \end{pmatrix}$. So $\mathrm{se}(\hat\beta_0) = \sqrt{0.88} = 0.938$ and $\mathrm{se}(\hat\beta_1) = \sqrt{0.08} = 0.283$, matching Exercise 10.2. The negative off-diagonal entry says the two estimates are negatively correlated: with all x positive, a steeper fitted slope forces a lower intercept.

Solution 11.2 (The hat matrix)**.**

(a) $\mathbf{H}^\top = \mathbf{X}\big((\mathbf{X}^\top\mathbf{X})^{-1}\big)^\top\mathbf{X}^\top = \mathbf{H}$, because $\mathbf{X}^\top\mathbf{X}$ is symmetric and so is its inverse. And $\mathbf{HH} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}(\mathbf{X}^\top\mathbf{X})(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top = \mathbf{H}$. Geometrically, $\mathbf{H}$ projects $\mathbf{y}$ onto the column space of $\mathbf{X}$, and projecting twice changes nothing.

(b) Using $\mathrm{trace}(\mathbf{AB}) = \mathrm{trace}(\mathbf{BA})$: $\mathrm{trace}(\mathbf{H}) = \mathrm{trace}\left((\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}\right) = \mathrm{trace}(\mathbf{I}_{p+1}) = p+1$. The average leverage is therefore $(p+1)/n$; a common rule of thumb flags points with leverage above two or three times this.

(c) For one predictor, $h_{ii} = 1/n + (x_i - \bar x)^2/\sum(x_j - \bar x)^2 = 0.2 + (x_i - 3)^2/10$, giving 0.6, 0.3, 0.2, 0.3, 0.6, which sum to $2 = p+1$. The two end points have the most leverage, three times that of the center point: extreme x-values pull the line hardest.

Solution 11.3 (Omitted-variable bias)**.**

(a) Substitute: $Y = 1 + 2X_1 + 3(0.5X_1 + U) + \varepsilon = 1 + 3.5X_1 + (3U + \varepsilon)$. The new error is uncorrelated with X_1, so the simple regression estimates 3.5: the direct effect 2 plus the indirect effect 3×0.5 that runs through X_2.

(b) It depends on the question. 2 is the partial effect: the change in Y when X_1 moves and X_2 is held fixed. 3.5 is the total association, which is the right answer if, when X_1 moves in reality, X_2 moves with it. Omitted-variable "bias" is bias relative to the partial effect. Neither number is causal unless the model's structure is itself causal, which the data cannot establish.

Solution 11.4 (Reading a multiple regression)**.**

(a) Holding gas price and load fixed, each additional GW of wind generation is associated with a \$0.30/MWh lower expected price.

(b) Wind is correlated with the omitted predictors. If windy days tend to be high-load winter days, or days with high gas prices, the simple regression credits wind with part of their positive effect on price. The partial and total associations can differ in sign (Simpson's paradox); the multiple regression separates them, provided the important confounders are included.

(c) $5.0 + 7.5(3) + 0.40(40) - 0.30(10) = 5.0 + 22.5 + 16.0 - 3.0 = \$40.50/\text{MWh}$.

Solution 11.5 (Adjusted R^2)**.**

(a) $\bar{R}^2 = 1 - (1 - R^2)\frac{n-1}{n-p-1}$. Three predictors: $1 - 0.60(49/46) = 0.361$. Five: $1 - 0.58(49/44) = 0.354$. Adjusted R^2 falls: the extra fit does not cover the degrees of freedom spent.

(b) $F = \dfrac{0.02/2}{0.58/44} = 0.76 < 3.21$. The two predictors add nothing detectable. (Dividing numerator and denominator of the usual nested F by SST gives this R^2 form.)

Solution 11.6 (The overall F-test)**.**

(a) Divide numerator and denominator of $F = \dfrac{\text{SSR}/p}{\text{SSE}/(n-p-1)}$ by SST, and use $\text{SSR/SST} = R^2$, $\text{SSE/SST} = 1 - R^2$.

(b) $F = \dfrac{0.40/3}{0.60/46} = 10.2 > 2.81$: the predictors jointly explain a significant share of the variation ($p \approx 3 \times 10^{-5}$).

Solution 11.7 (A block of predictors)**.** $F = \dfrac{(520 - 440)/2}{440/(60 - 5 - 1)} = \dfrac{40}{8.15} = 4.91 > 3.17$. Reject the hypothesis that both coefficients are zero ($p \approx 0.011$): the weather block belongs in the model, even if neither variable would be individually significant because the two are correlated.

Solution 11.8 (Variance inflation)**.** $\text{VIF} = 1/(1 - R_j^2)$ inflates the variance, so the standard error is inflated by $\sqrt{\text{VIF}}$:

R_j^2	VIF	SE inflation
0.50	2	1.41
0.80	5	2.24
0.90	10	3.16
0.99	100	10.0

At $R_j^2 = 0.9$ the standard error is three times what it would be with uncorrelated predictors, equivalent to having one-tenth of the data for estimating that coefficient. The usual thresholds of 5 and 10 correspond to standard errors inflated by factors of about 2.2 and 3.2.

Solution 11.9 (AIC versus BIC)**.** $n \ln(\text{SSE}/n)$ is $100 \ln 2.5 = 91.63$ for A and $100 \ln 2.3 = 83.29$ for B. AIC: A $= 91.63 + 6 = 97.63$; B $= 83.29 + 12 = 95.29$, so AIC prefers B. BIC ($\ln 100 = 4.605$): A $= 91.63 + 13.82 = 105.44$; B $= 83.29 + 27.63 = 110.92$, so BIC prefers A.

B's improvement in fit is worth 8.34 points. AIC charges 2 per extra parameter, so the three extra parameters cost 6 and B wins. BIC charges 4.6 each, so they cost 13.8 and A wins. The criteria disagree exactly when the extra parameters buy a moderate improvement; an out-of-sample comparison (Chapter 18) is the more direct test.

Solution 11.10 (Perfect collinearity)**.**

(a) $^\circ\mathrm{F} = 1.8\,^\circ\mathrm{C} + 32$ is an exact linear combination of the Celsius column and the intercept column. Infinitely many coefficient pairs give identical fits, so no unique estimate exists. Keep one temperature variable.

(b) The four indicators sum to one in every row, which equals the intercept column (the "dummy variable trap"). Drop one season, which becomes the baseline; each remaining coefficient is then the difference from that baseline. Alternatively, drop the intercept and keep all four.

Solution 11.11 (Jointly significant, individually not)**.**

(a) The two loads move almost in lockstep, so the data contain almost no information about the effect of one holding the other fixed. Each partial coefficient has a huge standard error (with $\rho = 0.97$, the VIF is about 17), although together they clearly explain the price.

(b) The fitted values and predictions can be perfectly good: collinearity affects how credit is divided between the predictors, not how well their combination fits. But the individual coefficients are unstable and cannot be interpreted.

(c) If the goal is prediction, possibly nothing. If the goal is interpretation, combine them (for example their average and their difference), drop one, or use regularization such as ridge regression (Chapter 18). Test the pair jointly rather than reading the individual t-statistics.

Chapter 12

When the Assumptions Break

Solution 12.1 (Reading residual plots)**.**

(a) Heteroskedasticity, error variance rising with the level. Use robust (White) standard errors; consider a log transformation of the outcome, or weighted least squares if the variance pattern is known.

(b) Wrong functional form: the relationship is curved. Add a squared term, transform a variable, or fit a nonlinear model; the coefficients of the straight line are biased.

(c) Positive autocorrelation. Test with Durbin–Watson or Breusch–Godfrey; use Newey–West (HAC) standard errors, or model the dynamics, for example by adding lagged variables.

(d) Heavy-tailed errors. In large samples inference is still roughly valid, but extreme residuals have outsized influence. Consider robust regression, a transformation, or bootstrap inference.

(e) A high-leverage point. A small residual does not mean it is harmless: it may have pulled the line onto itself. Compute its Cook's distance or refit without it to see how much the coefficients change (Exercise 12.6).

Solution 12.2 (Durbin–Watson by hand)**.**

(a) The successive differences are -0.2, 0.1, -0.6, -0.5, -0.4, -0.3, 0.2, 0.5, 0.6; their squares sum to 1.56. The squared residuals sum to 4.44. So $d = 1.56/4.44 = 0.35$.

(b) $\sum e_t e_{t-1} = 3.08$, so $r_1 = 3.08/4.44 = 0.69$, and $2(1 - r_1) = 0.61$. The approximation is rough here because with only ten residuals the end terms matter; it improves as n grows.

(c) d is far below 2: strong positive autocorrelation, visible to the eye as a slow swing from positive to negative and back. The OLS standard errors from this regression will be too small.

Solution 12.3 (What autocorrelation costs)**.**

(a) $(1 + 0.6)/(1 - 0.6) = 4$ for the variance, so 2 for the standard error.

(b) $200/4 = 50$: the 200 correlated observations carry about as much information about the mean as 50 independent ones.

(c) The standard error doubles to 0.20 and t halves to 1.25, no longer significant at any conventional level. The "finding" was produced by treating dependent observations as independent.

Solution 12.4 (The sign of omitted-variable bias)**.**

(a) Floor area raises price ($\beta_2 > 0$) and is positively related to bedrooms ($\delta > 0$): upward bias. The bedroom coefficient absorbs the value of space.

(b) Rain raises yield ($\beta_2 > 0$) but is negatively related to fertilizer use ($\delta < 0$): downward bias. Fertilizer looks less effective than it is, because it is used where conditions are worse.

(c) Age raises outages ($\beta_2 > 0$) and is positively related to maintenance ($\delta > 0$): upward bias. Maintenance may even appear to *cause* outages, a sign reversal produced entirely by confounding.

Solution 12.5 (Leverage)**.** $\bar{x} = 4$ and $\sum(x_j - \bar{x})^2 = 9 + 4 + 1 + 0 + 36 = 50$. Then $h = 0.2 + (x_i - 4)^2/50$: 0.38, 0.28, 0.22, 0.20, 0.92, summing to 2 as they must. The point at $x = 10$ has leverage 0.92, so its own y-value determines 92% of its fitted value. The line will pass very close to it whatever its y, which is why its residual will be small even if the point is wrong. With leverage this high, the estimated slope rests largely on a single observation.

Solution 12.6 (One point flips the slope)**.**

(a) Without it the slope is exactly 1. With it, $\hat{y} = 3.11 - 0.15x$: a single point reverses the sign of the slope, and R^2 falls from 1 to 0.06.

(b) The residuals are $-1.97, -0.82, 0.33, 1.48, 2.62, -1.64$. The largest is at $(5, 5)$, not at $(10, 0)$, because the influential point has pulled the line toward itself. A search for large residuals points at the wrong observation.

(c) Investigate the point before deciding anything. Is it a data error (a missing value coded as zero, a unit mistake)? If so, fix or remove it. If it is genuine, the relationship may not be linear over the whole range, and the conclusion should be reported with and without it. Do not delete it simply because it is inconvenient.

Solution 12.7 (Choosing a remedy for heteroskedasticity)**.**

(a) The coefficients remain unbiased and consistent; the usual standard errors are wrong, typically too small, so tests and intervals are unreliable.

(b) *Robust (White) standard errors*: keep OLS and fix the inference. Best as a default when the form of the heteroskedasticity is unknown and the sample is large. *Weighted least squares*: best when the variance structure is known or can be modeled well; it also restores efficiency. *Log transformation* of cost (and perhaps output): best when the spread grows with the level because effects are multiplicative, as is typical of costs; it often stabilizes the variance and makes coefficients interpretable as elasticities.

(c) Weight each observation by the inverse of its error variance: $w_i \propto 1/x_i^2$. Large firms, whose costs are noisier, count less.

Solution 12.8 (True or false)**.**

(a) False. It distorts the standard errors, not the coefficients.

(b) True.

(c) False in general. The CLT makes inference approximately valid in large samples if the error variance is finite; very heavy tails remain a concern.

(d) False. With $\delta = 0$ there is no bias, but the omitted variable's effect goes into the error, raising the residual variance and the standard errors.

(e) False. It costs precision (larger variances, fewer degrees of freedom), not bias.

(f) False. It adds noise to the error term, lowering precision and R^2, but leaves the slope unbiased.

(g) True for classical measurement error: attenuation bias (Exercise 12.9).

Solution 12.9 (Attenuation bias)**.**

(a) The OLS slope converges to $\mathrm{Cov}(Y, W)/\mathrm{Var}(W)$. Now $\mathrm{Cov}(Y, W) = \mathrm{Cov}(\beta X + \varepsilon,\ X + U) = \beta\,\mathrm{Var}(X)$ by independence, and $\mathrm{Var}(W) = \mathrm{Var}(X) + \mathrm{Var}(U)$. Divide.

(b) $2 \times 4/5 = 1.6$: a 20% understatement.

(c) The bias is in the probability limit: with infinite data the estimate converges to 1.6, not 2. The measured predictor is correlated with the error of the regression on W (that error contains $-\beta U$), which is a form of endogeneity. Remedies need outside information: a second, independent measurement of X (an instrument), or knowledge of $\mathrm{Var}(U)$ to correct the estimate.

Solution 12.10 (Delete the spike?)**.**

(a) Whether it is a data error: the correct market, units, and date, and whether the price was later revised. If it is an error, correct it.

(b) A genuine scarcity price is the most informative observation about how prices behave at high load. Deleting it produces a model that fits normal days well and is badly wrong exactly when it matters, a "comforting but false regularity." The real problem is probably the linear form: prices respond to load nonlinearly as the supply stack steepens.

(c) Model the nonlinearity (a convex function of load, or a regime for scarcity conditions); use a log or other transformation of price; use robust regression to limit the point's pull while keeping it; or analyze extreme days separately. In all cases, report results with and without the point so readers can see its effect.

Chapter 13

Analysis of Variance

Solution 13.1 (One-way ANOVA by hand)**.**

(a) Group means $5, 7, 9$; grand mean 7. SSB $= 4\left[(5-7)^2 + (7-7)^2 + (9-7)^2\right] = 32$. Within groups the squared deviations are $1, 0, 1, 0$ for A, $0, 1, 1, 0$ for B and $0, 1, 1, 0$ for C, so SSW $= 6$. SST $= 38 = 32 + 6$.

(b)

source	SS	df	MS	F
between	32	2	16.00	24.0
within	6	9	0.667	
total	38	11		

$F = 24.0 > 4.26$: reject equal means ($p = 0.0002$).

(c) SSB/SST $= 32/38 = 0.84$.

Solution 13.2 (Which groups differ?)**.** HSD $= 3.95\sqrt{0.667/4} = 3.95(0.408) = 1.61$. The differences are $|5-7| = 2$, $|7-9| = 2$ and $|5-9| = 4$, all larger than 1.61, so all three pairs differ, with familywise error controlled at 5%. The ordering is $A < B < C$.

Solution 13.3 (Completing a table)**.** Degrees of freedom: $k - 1 = 3$ between and $N - k = 20$ within. SSW $= 5 \times 20 = 100$, SST $= 160$, MSB $= 60/3 = 20$, $F = 20/5 = 4.0 > 3.10$: reject ($p \approx 0.02$). SSB/SST $= 0.375$.

Solution 13.4 (With two groups, $F = t^2$)**.**

(a) $s_p^2 = (2+2)/(4+4-2) = 0.667$, so se $= \sqrt{0.667(1/4 + 1/4)} = 0.577$ and $t = 4/0.577 = 6.93$.

(b) The grand mean is 7, so SSB $= 4(4) + 4(4) = 32$ on 1 degree of freedom, and MSW $= 4/6 = 0.667$. Then $F = 32/0.667 = 48.0 = 6.93^2$. With two groups ANOVA *is* the pooled two-sample t-test; ANOVA generalizes it to k groups.

Solution 13.5 (ANOVA as regression)**.**

(a) The fitted value for each group must be its group mean, so $\hat{y} = 5 + 2D_B + 4D_C$. The intercept is the mean of the baseline group A; each indicator coefficient is that group's difference from A.

(b) The regression's fitted values are the group means, so its SSR is the ANOVA's SSB and its SSE is SSW. Hence $R^2 = 32/38 = 0.84$, and the overall F on $(2, 9)$ degrees of freedom is the same 24.0. The t-tests on the two coefficients compare B and C with A, though without multiple-comparison control.

Solution 13.6 (Why not just run t-tests?)**.**

(a) $\binom{5}{2} = 10$.

(b) $1 - 0.95^{10} = 0.40$. (The pairwise tests are actually dependent, since they share groups, but the inflation is of this order.)

(c) $0.05/10 = 0.005$. With $k = 6$ there are 15 pairs, giving $1 - 0.95^{15} = 0.54$ uncorrected and a Bonferroni level of 0.0033. The number of pairs grows like $k^2/2$, which is why an omnibus F-test followed by controlled comparisons is the standard approach.

Solution 13.7 (An interaction)**.**

(a) Methods: A $(10 + 20)/2 = 15$, B $(18 + 12)/2 = 15$. Locations: North 14, South 16.

(b) Yes, a strong one. The effect of switching from A to B is $+8$ in the North and -8 in the South; the interaction contrast is $(10 - 20) - (18 - 12) = -16$. With no interaction the A–B difference would be the same in both locations.

(c) Wrong. On average over locations the methods tie, but in each location the choice matters a great deal: use B in the North and A in the South, gaining 8 MWh in each. When the interaction is strong, main effects average away the most useful information; report the cell means.

Solution 13.8 (When the assumptions fail)**.**

(a) Equal variances, badly: the largest standard deviation is nine times the smallest, and the group sizes are unequal, the combination in which the classical F-test is most distorted. Normality is also doubtful in the skewed group. Independence should be checked from the design.

(b) Welch's ANOVA, which does not assume equal variances. A transformation such as the logarithm, which may stabilize the variances and reduce the skewness at once (then the comparison is of geometric means). Or the Kruskal–Wallis test (Chapter 14), which compares groups through ranks, although it too is affected by very unequal spreads. A permutation test (Chapter 18) is a further option.

Chapter 14

Nonparametric Methods

Solution 14.1 (The sign test)**.**

(a) Binomial$(12,\, 1/2)$.

(b) $P(X \geq 10) = \left[\binom{12}{10} + \binom{12}{11} + \binom{12}{12}\right]/2^{12} = (66 + 12 + 1)/4096 = 0.019$. Two-sided, doubling by symmetry: 0.039. The new method wins significantly at 5% whichever alternative was specified in advance.

Solution 14.2 (The sign test with many pairs)**.** Under H_0 the count is Binomial$(100, 1/2)$, with mean 50 and standard deviation $\sqrt{100/4} = 5$. $z = (62 - 50)/5 = 2.4$, and the two-sided $p = 2(1 - 0.9918) = 0.016$.

Solution 14.3 (The Wilcoxon signed-rank test)**.**

(a) The absolute values in increasing order are $0.4, 0.7, 0.9, 1.2, 1.5, 2.1, 2.5, 3.0$, with ranks 1 to 8. The negative differences are -0.4 (rank 1) and -1.2 (rank 4), so $W_- = 5$ and $W_+ = 36 - 5 = 31$.

(b) $W_- \leq 5$ requires the negative ranks to be a subset of $\{1, \ldots, 8\}$ summing to at most 5: $\varnothing$, $\{1\}$, $\{2\}$, $\{3\}$, $\{4\}$, $\{5\}$, $\{1,2\}$, $\{1,3\}$, $\{1,4\}$, $\{2,3\}$, ten in all. One-sided $p = 10/256 = 0.039$; two-sided 0.078.

(c) The sign test sees only 6 positives out of 8, with two-sided $p = 2(1 + 8 + 28)/256 = 0.29$. The signed-rank test also uses the fact that the negative differences are small, giving $p = 0.078$, close to the t-test's 0.060. Using magnitudes (through ranks) recovers most of the information the sign test throws away.

Solution 14.4 (The Mann–Whitney test)**.**

(a) Pooled order: $12, 15, 18, 20, 22, 25, 27, 30, 31$, with ranks 1 to 9. Group A holds ranks $1, 2, 3, 5$, so $R_A = 11$ and $U_A = 11 - 10 = 1$. (U_A counts the pairs (a, b) with $a > b$: only $22 > 20$.)

(b) There are $\binom{9}{4} = 126$ equally likely rank sets. $U_A = 0$ needs ranks $\{1,2,3,4\}$ and $U_A = 1$ needs $\{1,2,3,5\}$, so $P(U_A \leq 1) = 2/126 = 0.016$ and the two-sided $p = 0.032$. Group A is significantly faster at the 5% level, a conclusion that requires no assumption about the shape of the response-time distribution.

Solution 14.5 (The Kruskal–Wallis test)**.** The pooled ranks are: G1: $2, 4, 1, 3$ (sum 10); G2: $6, 9, 5, 8$ (sum 28); G3: $10, 7, 11, 12$ (sum 40). The rank sums add to $78 = N(N+1)/2$, as they must. Then

$$H = \frac{12}{12 \cdot 13} \cdot \frac{10^2 + 28^2 + 40^2}{4} - 3(13) = \frac{12}{156}(621) - 39 = 8.77,$$

which exceeds 5.99 ($p \approx 0.012$). The groups do not share a common distribution; a follow-up rank comparison of pairs would locate the differences.

Solution 14.6 (One bad value)**.**

(a) The mean is -0.29, dragged below zero by -9.0, and the inflated standard deviation makes $t = -0.23$: no evidence of any change, and the estimated direction is wrong.

(b) Seven of the eight values are positive. The two-sided p is $2\left[\binom{8}{7}+\binom{8}{8}\right]/256 = 2(9)/256 = 0.070$, borderline, and the sign is right. The median, 0.85, describes the seven good values well.

(c) The sign test and the median, which are unaffected by the size of the bad value. The deeper lesson is to look at the data: a plot would have revealed the -9.0 at once.

Solution 14.7 (Choosing a test)**.**

(a) Mann–Whitney: ordinal data, two independent groups. (Expect many ties; software corrects for them.)

(b) Wilcoxon signed-rank: paired data, symmetric but heavy-tailed differences.

(c) Kruskal–Wallis, or ANOVA on log costs if the logs look roughly normal with similar spreads.

(d) Two-sample t-test: large samples of a roughly normal measurement, where the parametric test has the most power.

(e) Sign test: only the direction of each difference is reliable.

Solution 14.8 (The price of robustness)**.**

(a) Wilcoxon: about 105. Sign test: about 157. The signed-rank test gives up very little; the sign test gives up a lot.

(b) With heavy tails the mean and standard deviation, on which the t-test is built, are dominated by a few extreme values and become noisy, so the t-test loses power. Ranks cap the influence of any single observation. For distributions with heavy enough tails, the Wilcoxon test is more powerful than the t-test, sometimes by a wide margin, and it is never much worse. That asymmetry is the case for using it routinely when normality is in doubt.

Chapter 15

Time Series Analysis

Solution 15.1 (Stationary or not?)**.**

(a) Not stationary: the mean varies with the season. Remove the seasonal pattern (subtract a day-of-year average, or take differences at lag 365).

(b) Approximately stationary in mean. The variance is not constant in practice, since volatility clusters, so "stationary" holds only roughly; volatility models address this.

(c) Not stationary: it behaves approximately like a random walk. Take differences of logs, which gives (log) returns, case (b).

(d) Not stationary: trend plus seasonality, with fluctuations that probably grow with the level. Take logs, then a seasonal difference (lag 12) and possibly an ordinary difference.

(e) Stationary: this is white noise.

Solution 15.2 (How many differences?)**.**

(a) First differences: $3, 4, 3, 4, 3$. Second differences: $1, -1, 1, -1$.

(b) $d = 1$. The first differences fluctuate around a stable mean of about 3.4, with no remaining trend. Differencing again removes nothing useful and creates artificial negative autocorrelation: the second differences alternate in sign. Over-differencing is a real error, not a harmless precaution; it adds a non-invertible moving-average structure and inflates forecast variance.

Solution 15.3 (Autocorrelations by hand)**.** The mean is $45/9 = 5$, and the deviations are $0, 2, 3, 2, 0, -2, -3, -2, 0$, with $\sum d_t^2 = 34$. The lagged products sum to 24 at lag 1, 4 at lag 2, -12 at lag 3 and -17 at lag 4. Hence

$$r_1 = 0.71, \quad r_2 = 0.12, \quad r_3 = -0.35, \quad r_4 = -0.50.$$

Strong positive correlation at lag 1, falling through zero to strongly negative at lag 4: the signature of a cycle, here with a period of about 8 observations (high values are followed four steps later by low ones). With only nine observations these estimates are very noisy; the approximate standard error of each is $1/\sqrt{9} = 0.33$.

Solution 15.4 (The AR(1) model)**.**

(a) X_{t-1} depends only on past shocks, so it is independent of ε_t, and stationarity gives $\mathrm{Var}(X_t) = \mathrm{Var}(X_{t-1}) = v$. Then $v = 0.49v + 1$, so $v = 1/(1 - 0.49) = 1.96$.

(b) $\mathrm{Cov}(X_t, X_{t-1}) = \mathrm{Cov}(0.7X_{t-1} + \varepsilon_t,\ X_{t-1}) = 0.7v$, so the correlation is 0.7. Repeating the argument gives $\rho_k = 0.7^k$: $\rho_2 = 0.49$, $\rho_3 = 0.343$. The ACF decays geometrically.

(c) The forecasts are $0.7^h \times 2$: 1.40, 0.98, 0.686. The forecast error at horizon h is $\varepsilon_{T+h} + 0.7\varepsilon_{T+h-1} + \cdots + 0.7^{h-1}\varepsilon_{T+1}$, with variance $1 + 0.49 + \cdots + 0.49^{h-1}$: 1, 1.49, 1.73.

(d) The forecasts decay toward the mean 0, and the error variance rises toward the unconditional variance 1.96. Far enough ahead, the best forecast is the mean and the uncertainty is the unconditional spread: the predictable structure has been used up, but the intervals stay bounded because the series is stationary.

Solution 15.5 (The random walk)**.**

(a) $X_{T+h} = X_T + \varepsilon_{T+1} + \cdots + \varepsilon_{T+h}$, so the best forecast is X_T (the current value) and the error variance is h.

(b) $1.96\sqrt{h}$: 1.96, 3.92, 9.80, 19.6.

(c) The AR(1) forecast reverts to the mean and its uncertainty levels off at $\sqrt{1.96} = 1.4$. The random walk never reverts, and its forecast uncertainty grows without bound, like $\sqrt{h}$. Telling the two apart from a finite sample is hard when the AR coefficient is close to one, which is what unit-root tests try to do.

Solution 15.6 (The MA(1) model)**.** $\mathrm{Var}(X_t) = 1 + 0.25 = 1.25$. $\mathrm{Cov}(X_t, X_{t-1}) = \mathrm{Cov}(\varepsilon_t + 0.5\varepsilon_{t-1},\ \varepsilon_{t-1} + 0.5\varepsilon_{t-2}) = 0.5$, so $\rho_1 = 0.5/1.25 = 0.4$. Observations two or more periods apart share no shocks, so $\rho_k = 0$ for $k \geq 2$. The MA(1) ACF cuts off sharply after lag 1, while the AR(1) ACF decays geometrically and never reaches exactly zero. This difference is how the ACF helps identify the model.

Solution 15.7 (Reading ACF and PACF plots)**.**

(a) AR(2): an autoregression's PACF cuts off after its order.

(b) MA(1): a moving average's ACF cuts off after its order.

(c) Nonstationarity, probably a unit root. Difference the series ($d = 1$) and re-examine; confirm with an augmented Dickey–Fuller test.

(d) White noise: no linear structure to exploit. (Check also for nonlinear structure, such as volatility clustering in squared values.)

(e) Annual seasonality: use a seasonal difference or seasonal AR/MA terms at lag 12.

Solution 15.8 (Spurious regression)**.**

(a) About 76% (a typical run gives 75–77%), with a median R^2 of about 0.17, although the two series are unrelated by construction.

(b) About 5%, the nominal rate.

(c) Each random walk wanders, and two independent wandering series will often drift in the same or opposite directions over a finite sample, producing a fitted relationship. The OLS standard errors assume independent errors, but the residuals here are themselves a random walk, massively autocorrelated, so the t-statistics are hugely inflated. Differencing makes both series stationary and restores valid inference. Establish stationarity before regressing one time series on another.

Solution 15.9 (Checking a fitted model)**.**

(a) The residuals are not white noise: some short-run structure remains, so the model has left predictable information on the table and its forecast intervals are miscalibrated.

(b) A spike at lag 1 alone suggests adding an MA(1) term, giving ARIMA(1,1,1); alternatively ARIMA(0,1,1). Compare candidates by AIC or BIC, check that the new residuals look like white noise (for example with a Ljung–Box test), and, best of all, compare out-of-sample forecast errors on a held-out period.

Chapter 16

Logistic Regression and Generalized Linear Models

Solution 16.1 (Odds and log-odds)**.**

(a) Odds $0.2/0.8 = 0.25$; log-odds $\ln 0.25 = -1.386$.

(b) $p = 1/(1 + e^{-0.5}) = 0.622$.

(c) $p = \text{odds}/(1 + \text{odds}) = 3/4 = 0.75$.

(d) $p = 1/(1 + e^2) = 0.119$.

Solution 16.2 (A fitted logistic regression)**.**

(a) The log-odds are -2, 0 and 2, giving probabilities 0.119, 0.500 and 0.881.

(b) $e^{0.08} = 1.083$: each additional GW multiplies the odds of a spike by 1.083, an 8.3% increase in the odds (not in the probability). $e^{0.8} = 2.23$: each additional 10 GW multiplies the odds by 2.23.

(c) Where the log-odds are zero: $x = 4/0.08 = 50$ GW.

Solution 16.3 (Effects on the probability scale)**.**

(a) From 0.500 to 0.690, a rise of 0.19. From 0.961 to 0.982, a rise of only 0.021.

(b) At $p = 0.5$: $0.25(0.08) = 0.020$ per GW. At $p = 0.9$: $0.09(0.08) = 0.0072$ per GW.

(c) The same 10 GW increase multiplies the odds by the same factor 2.23 everywhere, but its effect on the probability depends on where you start: large near $p = 1/2$, small near 0 or 1. The odds ratio is the one number that describes the effect everywhere. When the probability effect is what matters for a decision, report it at specific, relevant values of the predictors.

Solution 16.4 (The linear probability model)**.**

(a) -0.1 and 1.1: impossible probabilities. (Even inside the data range, the line reaches 0 at $x = 10$ and 1 at $x = 60$, so it is at the edge of validity everywhere near the ends.)

(b) Unbounded predictions; non-normal errors (each error can take only two values); and built-in heteroskedasticity, since the error variance $p(1-p)$ changes with x. The linear probability model can still give reasonable average effects when probabilities stay well inside $(0, 1)$, but it is the wrong shape for a probability.

Solution 16.5 (Maximum likelihood by hand)**.**

(a) $\ell(p) = k \ln p + (n-k)\ln(1-p)$. Setting $\ell'(p) = k/p - (n-k)/(1-p) = 0$ gives $\hat{p} = k/n$; $\ell'' < 0$ confirms a maximum.

(b) The MLE of a function of a parameter is the function of the MLE, so $\hat{\beta}_0 = \ln \frac{\hat{p}}{1-\hat{p}} = \ln \frac{k}{n-k}$. With $k = 30$, $n = 120$: $\hat{p} = 0.25$ and $\hat{\beta}_0 = \ln(1/3) = -1.099$.

Solution 16.6 (Poisson regression)**.**

(a) $e^{0.5} = 1.65$ and $e^{1.5} = 4.48$ outages per month.

(b) $e^{0.2} = 1.22$: each additional year multiplies the expected outage rate by 1.22, a 22% increase.

(c) $e^{5\times 0.2} = e^1 = 2.72$. Effects multiply rather than add: five years of 22% increases compound to 172%, not 110%. (If the counts are overdispersed, the coefficients may be fine but the Poisson standard errors will be too small; Exercise 5.6.)

Solution 16.7 (A confusion matrix)**.**

(a) Accuracy $(40 + 120)/200 = 0.80$. Sensitivity $40/50 = 0.80$. Specificity $120/150 = 0.80$. Precision $40/70 = 0.57$.

(b) $150/200 = 0.75$: only five points below the classifier, without detecting a single spike. Accuracy alone says little when classes are unbalanced.

(c) Out of 10,000 days: 100 spikes, of which 80 are flagged; 9,900 normal days, of which 20%, or 1,980, are flagged. Precision $80/2{,}060 = 0.039$: fewer than 1 flag in 25 is a real spike. Accuracy is still 0.80, while "never flag" now scores 0.99. This is the base-rate effect of Example 3.1: sensitivity and specificity are properties of the classifier, but precision depends on how rare the event is.

Solution 16.8 (Thresholds and the ROC curve)**.**

(a) More days are flagged, so sensitivity rises (fewer spikes missed) and specificity falls (more false alarms). Where to set the threshold depends on the relative costs of the two errors, exactly as in choosing α (Exercise 8.2).

(b) 0.5: the model ranks spike and normal days no better than chance. 1.0: some threshold separates them perfectly.

(c) It depends only on how the scores *rank* the two kinds of days, not on where any cutoff is placed. It is in fact the Mann–Whitney statistic of Chapter 14, computed on the scores and rescaled to lie between 0 and 1.

Solution 16.9 (A likelihood ratio test)**.** The likelihood ratio statistic is the drop in deviance, 9.2, compared with χ^2_2 (two added parameters). Since $9.2 > 5.99$, reject the hypothesis that both coefficients are zero; $p = e^{-4.6} = 0.010$. This is the GLM counterpart of the nested F-test of Exercise 11.7.

Solution 16.10 (Three components)**.**

(a) Normal; identity link; $\beta_0 + \beta_1$ temp.

(b) Bernoulli; logit link; $\beta_0 + \beta_1$ leverage $+ \beta_2$ size.

(c) Poisson; log link; $\beta_0 + \beta_1$ age.

(d) Negative binomial; still a log link; same linear predictor. Only the random component changes, to one that allows variance above the mean. The linear predictor and link are the same as in (c).

Chapter 17

Bayesian Inference

Solution 17.1 (Beta–Binomial updating)**.**

(a) $\text{Beta}(2+9,\ 2+3) = \text{Beta}(11, 5)$, with mean $11/16 = 0.6875$.

(b) The prior mean is 0.5 and the observed proportion 0.75. With weight $n/(\alpha+\beta+n) = 12/16 = 0.75$ on the data and 0.25 on the prior: $0.75(0.75) + 0.25(0.5) = 0.6875$. The prior counts as $\alpha + \beta = 4$ pseudo-observations.

Solution 17.2 (How strong is the prior?)**.**

(a) $\text{Beta}(29, 23)$, with mean $29/52 = 0.558$. A prior worth 40 pseudo-observations dominates 12 real ones: the estimate moves only a quarter of the way from 0.5 toward 0.75.

(b) Flat: $\text{Beta}(901, 301)$, mean 0.7496. $\text{Beta}(20, 20)$: $\text{Beta}(920, 320)$, mean 0.7419. The two differ by less than 0.01.

(c) With little data the prior matters a lot; with much data the likelihood dominates and reasonable priors lead to nearly the same answer. When the prior is doing real work, say so and show how the answer depends on it (Exercise 17.9).

Solution 17.3 (One batch or two)**.**

(a) After the first batch: $\text{Beta}(1+3, 1+2) = \text{Beta}(4, 3)$. Using it as the new prior, after the second batch: $\text{Beta}(4+4, 3+1) = \text{Beta}(8, 4)$.

(b) All at once: 7 successes in 10 trials gives $\text{Beta}(1+7, 1+3) = \text{Beta}(8, 4)$, the same. In general the posterior after both batches is proportional to prior $\times$ likelihood$_1$ $\times$ likelihood$_2$; for independent data the likelihood of the combined data is the product of the two, so the order and grouping of the updates cannot matter. Yesterday's posterior is today's prior.

Solution 17.4 (A credible interval)**.**

(a) The 2.5th and 97.5th percentiles of $\text{Beta}(8, 4)$ give $(0.39,\ 0.89)$. $P(\theta > 0.5 \mid \text{data}) = 0.89$.

(b) $0.7 \pm 1.96(0.145) = (0.42,\ 0.98)$. With $n = 10$ the normal approximation is poor: the interval is forced to be symmetric about 0.7 and reaches nearly to 1, whereas the posterior is skewed and pulls the interval down.

(c) Credible interval: given the prior and the data, θ lies in $(0.39, 0.89)$ with probability 0.95. Confidence interval: the procedure that produced $(0.42, 0.98)$ covers the true θ in 95% of repeated samples (and, with $n = 10$, actually somewhat less, because the approximation is poor). Only the Bayesian statement is a probability about θ itself.

Solution 17.5 (Gamma–Poisson updating)**.**

(a) The likelihood is $\prod_i e^{-\lambda}\lambda^{x_i}/x_i! \propto \lambda^{\sum x_i} e^{-n\lambda}$. Multiplying by the prior $\lambda^{a-1}e^{-b\lambda}$ gives $\lambda^{a+\sum x_i - 1}e^{-(b+n)\lambda}$, the kernel of $\text{Gamma}(a + \sum x_i,\ b+n)$. The prior acts like b prior days with a prior events.

(b) $\sum x_i = 14$ and $n = 4$, so the posterior is $\text{Gamma}(18, 6)$ with mean 3.0. The prior mean is $4/2 = 2$ and the sample mean 3.5; with weight $n/(b+n) = 4/6$ on the data, $\frac{2}{3}(3.5) + \frac{1}{3}(2) = 3.0$.

(c) The 2.5th and 97.5th percentiles of $\text{Gamma}(18, 6)$ give $(1.78,\ 4.54)$.

Solution 17.6 (Normal prior, normal data)**.**

(a) Prior precision $1/100 = 0.01$; data precision $25/400 = 0.0625$; posterior precision 0.0725, so the posterior variance is 13.8 and the standard deviation 3.71. The weight on the data is $0.0625/0.0725 = 0.86$, and the posterior mean is $0.14(50) + 0.86(60) = 58.6$.

(b) $58.6 \pm 1.96(3.71) = (51.3,\ 65.9)$. The frequentist interval is $60 \pm 1.96(4) = (52.2,\ 67.8)$. The Bayesian interval is shifted toward the prior mean and is a little narrower, because the prior adds information worth $400/100 = 4$ observations.

Solution 17.7 (Fair or biased?)**.**

(a) Likelihoods (the binomial coefficient $\binom{10}{8} = 45$ cancels, but is included here): fair, $45(0.5)^{10} = 0.0439$; biased, $45(0.7)^8(0.3)^2 = 0.2335$. Posterior $P(\text{biased}) = 0.2335/(0.2335 + 0.0439) = 0.842$.

(b) $0.2335/0.0439 = 5.31$: the data are 5.3 times as likely under the biased hypothesis.

(c) Posterior odds = Bayes factor $\times$ prior odds $= 5.31 \times (0.1/0.9) = 0.59$, so $P(\text{biased}) = 0.59/1.59 = 0.37$. The same evidence leads to different conclusions from different starting points. The Bayes factor is the part everyone can agree on.

Solution 17.8 (Which statement is legitimate?)**.** (a) is justified: it is the definition of a credible interval. (b) is not: a confidence interval supports no probability statement about θ. (c) is justified (approximately, since the normal approximation is weak for small n). (d) is not: the credible interval depends on the prior as well as the data, and that dependence should be stated, not hidden.

Solution 17.9 (Reporting prior sensitivity)**.** Beta$(3, 39)$: mean $3/42 = 0.071$. Beta$(3, 57)$: mean $3/60 = 0.050$. Beta$(3, 137)$: mean $3/140 = 0.021$. The observed rate is 0.05.

With only two defaults, the answer ranges from about 2% to 7% depending on the prior, so the prior is doing real work. Report all three (or a range of priors), explain the reasoning behind each, and say plainly that the data alone cannot settle the question. More data, or a credible prior built from comparable contracts, would narrow it.

Chapter 18

Resampling and Modern Computational Methods

Solution 18.1 (Bootstrap mechanics)**.**

(a) Each of the three draws can be any of the 3 values: $3^3 = 27$ ordered resamples, all equally likely. Ignoring order, the number of multisets of size 3 from 3 values is $\binom{5}{3} = 10$ (they are not equally likely: $\{1,3,8\}$ arises in 6 orders, $\{1,1,1\}$ in only one).

(b) Each draw avoids 8 with probability $2/3$: $(2/3)^3 = 8/27 = 0.30$.

(c) Each of the n independent draws misses the given observation with probability $1 - 1/n$. For $n = 10$: 0.349; for $n = 100$: 0.366; the limit is $e^{-1} = 0.368$. So a typical resample contains about 63% of the distinct original observations, some of them repeated.

Solution 18.2 (Bootstrapping a median)**.**

(a) The median is 11.9 and the mean 13.41; the two long repairs (30.5 and 22.4) pull the mean up.

(b) With 100,000 resamples, the bootstrap standard error of the median is about 1.0 and the 95% percentile interval about $(9.9,\ 14.2)$. Different random seeds give slightly different answers, for example standard errors between 0.97 and 1.02 with 2,000 resamples. The bootstrap distribution of a median is lumpy, because a resampled median from an odd-sized sample must equal one of the original values, so interval endpoints jump between data values.

(c) The bootstrap standard error of the mean is about 1.46; the formula gives $s/\sqrt{n} = 5.86/\sqrt{15} = 1.51$. The small difference is explained in Exercise 18.3. The median has a smaller standard error than the mean here because the two extreme values inflate the variability of the mean but barely affect the median.

Solution 18.3 (The bootstrap standard error of a mean)**.**

(a) X_i^* takes each value x_j with probability $1/n$, so its mean is $\frac{1}{n}\sum x_j = \bar{x}$ and its variance $\frac{1}{n}\sum(x_j - \bar{x})^2 = \hat{\sigma}^2$. The bootstrap treats the sample as the population, and the variance of that population divides by n.

(b) The draws are independent, so $\mathrm{Var}(\bar{X}^*) = \hat{\sigma}^2/n$. Since $\hat{\sigma}^2 = \frac{n-1}{n}s^2$, the standard error is $\sqrt{(n-1)/n}\, s/\sqrt{n}$.

(c) $\sqrt{14/15} \times 1.513 = 1.461$, which matches the simulated 1.46 of Exercise 18.2. The bootstrap standard error is slightly too small in small samples, by the same factor that Bessel's correction fixes.

Solution 18.4 (A permutation test by hand)**.**

(a) $26/3 - 3 = 5.67$.

(b) The ten possible differences, sorted, are -5.17, -3.50, -2.67, -1.00, -0.17, 0.67, 1.50, 1.50, 3.17, 5.67. For example, $A = \{9, 10\}$ gives $\bar{x}_B - \bar{x}_A = 13/3 - 9.5 = -5.17$.

(c) Only the observed labeling reaches 5.67, so $p = 1/10 = 0.10$. That is also the smallest p-value possible with groups of 2 and 3: however strong the effect, the test cannot reject at 5%. Exact tests on tiny samples have coarse null distributions, and the sample size limits what can ever be concluded.

Solution 18.5 (k-fold cross-validation)**.**

(a) Each fit trains on 80 and tests on 20; there are 5 fits. Leave-one-out needs 100 fits, each training on 99.

(b) The average, 4.3 (equal fold sizes, so a simple average).

(c) The training error is computed on data the model was fit to, so it includes the noise the model has learned and is optimistic. The cross-validated error is computed on data each fit has not seen, and is the honest estimate of error on new data: report 4.3. A large gap between the two is itself a sign of overfitting.

Solution 18.6 (Ridge and lasso in one dimension)**.**

(a) $\hat{\beta} = 50/25 = 2$.

(b) The derivative is $-2\sum x_i(y_i - \beta x_i) + 2\lambda\beta = 0$, giving $\beta(\sum x_i^2 + \lambda) = \sum x_i y_i$. For $\lambda = 25$: $50/50 = 1$. Ridge shrinks the estimate proportionally, but never all the way to zero.

(c) For $\beta > 0$ the penalty is $\lambda\beta$, and the derivative $-2\sum x_i y_i + 2\beta\sum x_i^2 + \lambda = 0$ gives the stated formula, valid while it is positive, that is, while $\lambda < 2\sum x_i y_i = 100$. For $\lambda \geq 100$ the objective is increasing in $|\beta|$ on both sides of zero, so the minimum is at exactly $\beta = 0$. For $\lambda = 40$: $(50 - 20)/25 = 1.2$. For $\lambda = 100$: 0. Lasso subtracts a fixed amount (soft-thresholding) and sets small coefficients exactly to zero, which is how it performs variable selection.

Solution 18.7 (Choosing the penalty)**.**

(a) $\lambda = 1$, which minimizes the cross-validated error. Training error always falls as the penalty is relaxed, because the fit is less constrained, so it cannot be used to choose λ; at $\lambda = 0$ (ordinary least squares) the model is fitting noise, as the gap of 2.7 between training and validation error shows.

(b) Training MSE rises steadily with λ: more shrinkage, more bias, worse in-sample fit. Cross-validated MSE is U-shaped: small λ gives low bias but high variance (overfitting), large λ gives low variance but high bias (underfitting), and the minimum balances the two. This is the bias–variance decomposition of Chapter 7 traced out empirically.

Solution 18.8 (Leakage)**.**

(a) Yes, often dramatically. The predictors were chosen because they happen to correlate with the outcome in all 200 observations, including those later used as test folds. The test folds are no longer unseen: information about them leaked into the model through the selection step. The cross-validated error will suggest real predictive power where none exists.

(b) Every data-dependent step, including variable selection, standardization fitted to the data, and tuning of λ, must be repeated inside each training fold, using only that fold's training data. Then the test fold is genuinely new. Done this way, cross-validation correctly reports no predictive power.

Solution 18.9 (When the bootstrap fails)**.**

(a) The bootstrap treats the sample as a stand-in for the population. A biased sample stays biased under resampling: the bootstrap faithfully quantifies the variability of an estimate of the wrong quantity.

(b) A resample's maximum equals the original maximum whenever the largest observation is drawn at least once, with probability $1 - (1 - 1/n)^n \approx 0.63$. The bootstrap distribution therefore puts about 63% of its mass on a single value and can never exceed the observed maximum, while the true sampling distribution of the maximum is continuous and often lies above it. Extremes are exactly where the sample has least information.

(c) Resampling individual days destroys the time ordering and the dependence, so the bootstrap treats the days as independent and understates the variability of statistics such as the mean (compare Exercise 12.3). A block bootstrap, which resamples consecutive blocks of days, preserves short-range dependence.